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Control of dynamic ride-hailing networks with a mixed fleet of autonomous vehicles and for-hire human drivers

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ABSTRACT

This study examines a ride-hailing platform that employs a mixed fleet of autonomous vehicles (AVs) and human drivers to offer on-demand ride-hailing services across a network over a finite planning horizon. In the mixed fleet, AVs have a fixed fleet size and are centrally managed by the platform, while human drivers can dynamically enter and exit from the ride-hailing market based on the prospective earning opportunities at various time periods. We introduce a game-theoretic model that encapsulates the strategic behavior of human drivers, passengers, and the ride-hailing platform. In this model, passengers choose between ride-hailing and alternative transportation options by considering fares and waiting times. Human drivers schedule their working hours in response to potential earnings and strategically relocate to maximize passenger pickups, while the platform adjusts pricing and AV repositioning strategies to maximize its profit. The solution is characterized as a Nash equilibrium within a two-player game, with one player representing the ride-hailing platform, and the other as a virtual representative of the collective human drivers. To address the non-convex nature of the game, we employ a convex relaxation technique to ascertain a near-optimal solution framed as an ϵ-Nash equilibrium, with ϵ accurately characterized. The proposed model and solution algorithm are validated in a case study for Manhattan, New York City. The simulation results underscore that the distribution of human drivers is closely aligned with the immediate, minute-to-minute fluctuations in passenger demand, enabling them to effectively meet the surge during peak demand windows. Conversely, the deployment patterns of AVs are more attuned to the extended, daily cycles of passenger movements throughout different city zones, from daytime to nighttime. We further compare the mixed fleet model with scenarios where the platform exclusively utilizes human drivers or AVs, respectively. The comparison show that the mixed fleet model excels in balancing the supply and demand over space and time, thereby leading to shorter wait times and reduced travel cost for ride-hailing passengers, as well as improved profit for the ride-hailing platform.

1. Introduction

Transportation network companies (TNCs), such as Uber, Lyft, and Didi, have significantly transformed urban mobility in recent years (Zhong et al., 2022; Sun and Ertz, 2021). They provide on-demand ride-hailing services that effectively meet passengers' travel needs at any time and place (Liu et al., 2022; Feng et al., 2021; Xu et al., 2021), with such convenience contributing to their

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growing user base. While traditional TNC platforms primarily depend on human drivers, recent advances in autonomous-driving technology have facilitated the integration of AVs into these platforms for on-demand ride-hailing services, which is typically referred to as autonomous mobility-on-demand (AMoD) services. This progression has attracted the attention of numerous corporations, including Waymo, NVIDIA, Zoox, and AutoX, fueling increased competition in the AMoD market. Currently, global investment in AV technology has topped $200 billion.

Despite concerted efforts, developing Level-5 fully autonomous vehicles for widespread commercial use remains a significant challenge, as they must safely handle all types of driving scenarios, which is extremely difficult due to the vast range of potential corner cases the vehicles may encounter. In contrast, Level-4 AVs operated within geofenced areas under favorable weather conditions have already been successfully deployed in various cities, such as Phoenix, San Francisco, and Shenzhen (Waymo, 2020). In the near future, a more practical scenario is that Level-4 AVs be jointly managed with a group of for-hire human drivers, forming a mixed fleet to offer ubiquitous on-demand ride-hailing services to TNC passengers at all times and locations. Compared to a traditional TNC platform that relies solely on human drivers, AV-enabled mobility services present significant advantages by reducing the platform's reliance on for-hire human drivers and improving its profitability (Wang et al., 2020). More importantly, AVs and human drivers can be spatially and temporally complementary to each other when they are jointly managed on a TNC platform. From the spatial dimension, AVs can be centrally dispatched by the platform to relocate to preferable locations, complementing the self-interested human drivers and enabling the platform to mitigate the spatial mismatch between supply and demand more efficiently at a lower cost. From the temporal dimension, AVs can serve as the primary means of transportation for base demand due to their lower variable costs, while human drivers can help meet peak demand since they are gig workers with more flexible work schedules (i.e., only paying for working hours) (Freund et al., 2022). In this case, the joint management of these two can better address the time variability of travel demand. Therefore, employing a mixture of human drivers and AVs offers numerous benefits to TNC platforms, enabling them to provide high-quality mobility services while ensuring financial sustainability.

When AVs are mixed with for-hire human drivers to provide on-demand ride-hailing services, the modelings and control of the TNC platform become inherently complex. In the mixed fleet, AVs are fully controlled and dispatched by the platform, whereas human drivers not only make their own working schedules (i.e., the time to enter and leave the platform) based on the temporal distribution of prospective earning opportunities in the long term (e.g., one day) but also make repositioning decisions[1] based on the spatial distribution of potential demand in the short term (e.g., 10 min). These short-term and long-term decisions of the human drivers involve both spatial (e.g., repositioning locations) and temporal (e.g., work schedule) elements, occurring over distinct time spans. They are also intimately entangled with the dynamics and spatial distribution of the AV fleet, as well as passenger demand. Therefore, to fully understand the optimal operational decisions of the TNC platform with mixed fleet, it is essential to capture the incentives of human drivers at multiple time scales and characterize its interaction with AV fleets, ride-hailing passengers, and the TNC platform over both space and time. However, in the existing literature, we have not found any work that considered the joint management of human drivers and AVs while simultaneously capturing the spatial (i.e., network) and temporal (i.e., dynamics) perspectives of the mixed fleet in the ride-hailing market.

To address this research gap, we investigate a platform utilizing a combination of AVs and human drivers to provide on-demand ride-hailing services over a transportation network within a finite horizon. In the mixed fleet, AVs have a fixed fleet size and are centrally managed by the TNC platform, while human drivers can dynamically enter and exit from the ride-hailing market based on their perceived prospective earning opportunities at various time periods. They also strategically reposition themselves between rides to optimize the likelihood of encountering their next passenger. The dynamics and spatial distribution of the mixed fleet influence passengers' incentives as well as the TNC platform's optimal control decisions. We will characterize the strategic interactions between human drivers, passengers, and the TNC platform across space and time, revealing the spatial and temporal synergy between AVs and human drivers in the ride-hailing market. The main contributions of this paper are summarized as follows:

(1) We develop a game-theoretic dynamic network model to characterize the strategic interaction between human drivers, ride-hailing passengers, and the TNC platform. In the proposed model, passengers make mode choices between TNC and other alternative transport options based on service fare and waiting time; human drivers determine their work schedules based on perceived prospective earnings at various time periods and make strategic repositioning decisions between rides to maximize their chance of meeting the next passenger; the TNC platform optimizes spatiotemporal pricing strategies and AV repositioning policies to maximize profit. The model's solution is characterized as the Nash equilibrium of a two-player game, where one player represents the TNC platform, and the other player is a virtual representative of the collective human drivers who make self-interested scheduling decisions. *To the best of our knowledge, we are among the first to propose a dynamic network model that simultaneously characterizes the spatial and temporal perspectives of the mixed fleet on a TNC platform.*

(2) The solution to the formulated two-player game is solved using a best response algorithm, where each player iteratively computes the optimal decisions in response to the other player's given decisions until convergence. However, since the platform's profit maximization problem is non-convex, verifying whether the candidate solution derived from the best response algorithm is indeed a Nash equilibrium remains challenging, as the TNC platform's operational decisions may be a local solution to the non-convex profit-maximization problem. To address this challenge, we propose a novel procedure for evaluating the optimality of the candidate Nash equilibrium based on a convex relaxation for the profit-maximization problem,

[1] Note that vehicle relocation in the ride-hailing market typically refers to strategically moving or reallocating vehicles to different areas or locations where there is anticipated demand or to better serve the needs of the market. This is a crucial aspect for ride-hailing companies to maximize efficiency and reduce wait times for customers.

which provides an upper bound on the distance between the objective value at the candidate solution and the globally optimal solution. This establishes the candidate solution derived from the best response algorithm as an ϵ-Nash equilibrium, which can be viewed as an approximation of the exact Nash equilibrium, with the approximation error quantified by ϵ. Note that the value of ϵ is rigorously characterized by the proposed algorithm.

(3) The model and solution techniques we developed are validated with data from Manhattan, New York City. The simulation studies underscore that the distribution of human drivers is closely aligned with the immediate, minute-to-minute fluctuations in passenger demand, enabling them to effectively meet the surge during peak demand windows. Conversely, the deployment patterns of AVs are more attuned to the extended, daily cycles of passenger movements throughout different city zones, from daytime to nighttime. Furthermore, we compare the mixed fleet model with two benchmarks, including: (a) A platform operating exclusively with for-hire human drivers; and (b) A platform operating exclusively with AVs. We find that compared to the benchmark cases, the mixed fleet model capitalizes on the complementary strengths of human drivers and AVs to achieve a better balance between supply and demand throughout various times and locations, which translates into shorter waiting times and reduced travel cost for passengers, as well as improved profit for the platform. This underscores the strategic benefit of integrating AVs with human drivers, optimizing the ride-hailing platform's operational efficiency and enhancing customer satisfaction.

2. Related works

2.1. Management of TNC platforms

Pricing is a recurring topic in the modeling and optimization of TNC platforms, studied from both temporal and spatial perspectives. From the temporal perspective, Guda and Subramanian (2019) investigates dynamic pricing to regulate demand fluctuations across market locations (zones), demonstrating its effectiveness when supply exceeds demand in specific zones. Cachon et al. (2017) reports that both providers and consumers benefit from temporal pricing due to improved provider utilization, lower customer prices during normal demand, and broader service accessibility during peak demand. In contrast, Banerjee et al. (2015) finds that dynamic pricing yields better robust performance with system parameters, while only marginally improving TNC platform profits. Nourinejad and Ramezani (2020) designs a dynamic non-equilibrium model to capture changes in passengers, idle vehicles, and occupied vehicles, exploring three types of pricing policies. The results indicate that TNC platforms can offer higher wages to drivers than revenues from riders, potentially benefiting overall profits when trip demand increases and supplied vehicles decrease simultaneously. Apart from dynamic pricing, spatial pricing has been extensively studied in the context of the ride-sourcing market. Bimpikis et al. (2019) reveals that maximized profits and consumer surplus at equilibrium align with optimal spatial pricing and compensation policies under balanced trip demand conditions on the network. Afifah and Guo (2022) examines spatial pricing in a Stackelberg framework to minimize locational imbalance. Chen et al. (2019) investigates an anisotropic spatial pricing strategy based on an analytical bi-level optimization model, wherein the platform establishes a spatially heterogeneous pricing strategy to maximize total profit at the upper level while solving the trip distribution model at the lower level. They claim that profit in a specific area can be influenced by trip demands from other areas. Besbes et al. (2021) contends that spatial pricing policies can encourage human drivers to reposition towards more profitable regions for the platform. Li et al. (2021) examines spatial pricing with a congestion charge on a ride-sourcing platform, showing that congestion can be effectively mitigated by a one-directional cordon charge and revenue-raising can be achieved by a trip-based congestion charge. Furthermore, Chen et al. (2021) develops a reinforcement learning-enhanced agent-based modeling and simulation system to tackle the problem of spatial–temporal pricing for a ride-sourcing platform.

In addition to pricing, repositioning of idle TNC drivers is another important aspect widely studied in the relevant literature. For instance, Li et al. (2021) proposes a probability model that captures potential earnings in different zones to characterize the repositioning flow of human drivers in the ride-hailing market. Beojone and Geroliminis (2021) assumes that idle drivers prefer to move to the closest high-demand areas, then randomly circulate in this region awaiting their next assignment. Dong et al. (2022) investigates driver repositioning strategy under a dual sourcing labor supply, where freelance drivers search for customers following their patterns, while contracted drivers reposition in a centralized manner to narrow spatial demand-supply gaps. Guo et al. (2021) employs a robust optimization framework to incorporate passenger-driver matching into vehicle rebalancing problems, producing improved rebalancing strategies. Karamanis et al. (2021) formulates a non-linear cost flow problem accounting for the supply–demand relationship of rides to address vehicle redistribution in ride-sourcing markets. Jiao et al. (2021) develops a batch training algorithm that yields positive results compared to experienced drivers who perform idle-time repositioning based on their expertise. Yang and Ramezani (2022) introduces a learning-based repositioning method in ride-hailing systems, considering reactions to short-term trip demand and long-term platform efficiency (e.g., order cancellation rates and passenger waiting times). Shou and Di (2020) focuses on a mean-field multi-agent reinforcement learning approach to tackle unserviced passenger requests induced by vacant for-hire drivers' cruising behavior during the passenger-seeking process. Qian et al. (2022) proposes a deep relocating option policy leading human driver agents to avoid staying in oversupplied areas and reposition to underserved areas. *We emphasize that the aforementioned works primarily focus on the management of human driver fleets, neglecting the interplay between human drivers and autonomous vehicles in the ride-hailing market.*

2.2. AMoD in TNC platforms

Another stream of research focuses on the modeling and control of AVs for autonomous mobility-on-demand (AMoD) systems. For example, Tsao et al. (2018) proposes a stochastic model predictive control (MPC) algorithm incorporating short-term probabilistic prediction for dispatching and repositioning AVs in an AMoD system. Similarly, Tsao et al. (2019) presents a MPC algorithm for optimizing ride-sharing AMoD system routes, thereby enhancing social welfare. Lai and Li (2024) investigated spatiotemporal pricing and fleet management for AMoD systems while taking elastic demand into account, where an integrated decomposition and dynamic programming approach was proposed to address the platform optimization problem. Iglesias et al. (2018) introduces a data-driven, end-to-end framework for computing the rebalancing strategy and minimum feasible fleet size in an AMoD system. Wen et al. (2019) evaluates the importance of demand information for AMoD systems, while Wollenstein-Betech et al. (2021) investigates route-planning policies for intermodal AMoD systems, taking congestion into account. Numerical simulations indicate that integrating AMoD with public transit significantly improves overall transportation system performance. Chen and Liu (2023) explores an integrated optimization problem for an electric AMoD system, addressing long-term charging facility deployment and short-term vehicle assignment, relocation, and charging decisions. Hyland and Mahmassani (2020) conducts a quantitative analysis of ride-sharing AMoD system operations, finding that pickup time negatively impacts overall operational performance by affecting travel and waiting times. For a comprehensive review of AMoD system control and operation, please refer to Zardini et al. (2022) and Narayanan et al. (2020). In addition to the aforementioned works, AMoD has also been studied in the context of autonomous and electric vehicles. For instance, Rossi et al. (2019) extends the AMoD system to incorporate the electric power network, considering vehicle charging and battery depreciation. The findings underscore the economic and societal value of AMoD and its interaction with the power network. Iacobucci et al. (2019) integrates vehicle-to-grid (V2G) charging with optimized routing choice and relocation decisions in an AMoD system, demonstrating that the system can reduce charging costs without compromising passenger waiting times by considering different timescales for charging (long-term) and routing and relocation (short-term). Ding et al. (2021) combines the AMoD fleet with battery swapping stations to determine swapping schedules and vehicle rebalancing. Paparella et al. (2022) optimizes fleet size, vehicle battery capacity, and operation strategy for electric AMoD systems to identify a global solution. *Note that all the aforementioned works focus on AV fleets, while the interaction between AVs and human drivers on ride-hailing platforms is not explicitly considered.*

2.3. Mixed fleet in ride-hailing systems

Several studies have examined the interactions between AVs and human-driven vehicles when they are jointly managed to provide mobility-on-demand networks. For example, Wei et al. (2019, 2020) analyze the interaction between human-driven vehicles and AVs within a multi-location equidistant network to optimize platform profit. The results suggest that the platform may prefer a mix of AVs and human drivers to maximize profit in certain cases. Yang et al. (2020, 2021) employ a Stackelberg game model to represent an AMoD system with a mixed fleet, in which the platform serves as the leader and human drivers as followers. A steady-state model is proposed for determining planning variables, and a time-varying model is formulated for designing a real-time coordination algorithm for AVs. Yao et al. (2020) investigates the influence of AVs on the hybrid ride-hailing market with human drivers using data-driven agent-based modeling and simulation. The simulation reveals that passenger waiting time can be significantly reduced by employing a small portion of AVs. Mo et al. (2022) studies a monopoly transportation network company (TNC) market with mixed fleets of AVs and human drivers, characterizing the impacts on traffic congestion through an aggregated market equilibrium model. Similarly, Castro and Frazelle (2022) proposes a game-theoretic model of a ride-hailing platform under mixed autonomy, with the results indicating that the number of AVs is essential for tempering competition for rides and achieving a profitable balance between two labor sources. Castro et al. (2023) further examines how the introduction of autonomous vehicles may lead to spatial inequality and deteriorate service levels. Xie et al. (2023) develops a multi-agent reinforcement learning framework for managing dynamic MoD platforms with mixed fleets, addressing the interaction between the platform and self-interested human drivers through a two-sided design. Siddiq and Taylor (2022) elucidates the impacts of AV ownership (i.e., platform-owned and individually-owned) on platform profit and social welfare, showing that AV cost also interacts with social welfare and profit. Wang et al. (2024) examines the hybrid planning and operation strategies for a ride-hailing network that runs a mixture of level-4 AVs and human drivers. The case studies show that the passengers in remote zones undergo higher trip costs with the decreasing cost of AV. Wang and Li (2024) investigates the competition between autonomous and traditional ride-hailing platforms and examines the impacts of AVs on traditional ride-hailing market under platform competition. Regarding regulation policy in the ride-hailing market under mixed autonomy, Ao et al. (2024) proposes a minimum wage policy for human drivers and geo-fencing for AVs to regulate the TNC market with mixed autonomy. Chen et al. (2023) investigates operating strategies and various government regulations on the TNC platform with a mixed fleet. Equilibrium analysis reveals that social welfare and platform profit can be improved with low maintenance costs for AVs.

Our paper is distinct from all aforementioned works as we jointly model the temporal and spatial interactions between AVs and human drivers for ride-hailing services while considering both human driver scheduling decisions and repositioning decisions. These strategic decisions of drivers not only occur over space and time, across distinct time granularity, but also influence the dynamics and spatial distribution of AVs, thus understanding these spatiotemporal interactions is crucial for harnessing the synergy between AVs and human drivers for ride-hailing services. However, this remains under-examined in all existing literature.

3. Mathematical model

Consider a ride-hailing platform that employs a mixed fleet of AVs and human drivers to offer mobility-on-demand services (MoD)[2] over a transportation network. Let $\mathcal{N} = \{1, 2, \ldots, N\}$ denote the set of zones, where N is the total number of zones in the city. Each trip request starts from an origin zone $i \in \mathcal{V}$ and ends in a destination zone $j \in \mathcal{V}$, taking an average travel time of t_{ij}, which is assumed to be exogenous and independent of the traffic flow of ride-hailing vehicles. To formulate the dynamics of the ride-hailing network, a planning horizon of length T is considered, which can be divided into T consecutive time intervals, each of length ΔT. For notation convenience, let $\mathcal{T} = \{1, 2, 3, \ldots, T\}$ represent the set of indexes for time intervals. We consider the case where the ride-hailing platform makes control decisions for the entire horizon to maximize its accumulative profit, while passengers, AVs, and human drivers interact with each other over space and time to achieve their respective objectives. The rest of this section presents a mathematical formulation to characterize these interactions.

3.1. Incentives of passengers

Passengers are faced with a multitude of travel choices, ranging from ride-hailing to public transportation (including buses and railways) and various other alternatives. Each transportation mode incurs distinct travel costs, which will influence the selection of travel mode of passengers. To model this selection process, we have defined a generalized travel cost for ride-hailing services. This cost, applicable for travel from zone i to zone j during the time interval t by ride-hailing services, is calculated as a weighted sum that factors in the waiting time for pickup, the time spent en-route, and the fare of the ride. The particular form of the cost can be articulated as follows:

$$c_{ij}^t = \alpha \cdot w_{i,t}^p + \beta \cdot t_{ij}^t + r_{ij}^t \cdot d_{ij}, \qquad i, j \in \mathcal{N}, t \in \mathcal{T}. \tag{1}$$

where $w_{i,t}^p$ denotes the average waiting time for passengers to be picked up by idle ride-hailing vehicles in zone i at time $t \in \mathcal{T}$; t_{ij}^t denotes the average en-route time for a trip from zone i to zone j starting at time t[3]; Therefore, t_{ij}^t represents the average travel time, which is endogenously determined by the likelihood of being allocated to an AV or human driver. We will provide a more detailed characterization of t_{ij}^t subsequently. d_{ij} is the average distance of ride-hailing trips from zone i to zone j; r_{ij}^t is average per-distance ride fare for ride-hailing trip from zone i to zone j at time t; α represents the value of time for the passenger when she/he is waiting to be picked up by a ride-hailing vehicle; β is the value of time for the passenger when she/he is on-board and traveling on the road. Since passengers are typically less patient while waiting to be picked up (Mohring et al., 1987) compared to while traveling en-route, we assume that $\alpha > \beta$. Moreover, the generalized cost model introduced in (1) presumes that the fares for ride-hailing trips completed by AVs and those by human drivers are the same, and that passengers cannot choose between being transported by an AV or a human driver. We believe these assumptions are justified because assigning passengers to a particular type of idle vehicle may significantly increase their wait time, thus raising the costs for those who prefer to choose a certain vehicle type for their ride-hailing trip, which may deter them from selecting this option.

The passenger waiting time $w_{i,t}^p$ is an endogenous variable that depends on the supply and demand of the ride-hailing market. We suppose that each ride request is matched to the closest idle vehicle regardless of its vehicle type. Furthermore, we assume that each passenger originating from zone i is only matched to the idle vehicles in the same zone.[4] Let $N_i^{I,t}$ represent the total number of idle vehicles in zone i at time interval t, which includes both idle AVs and idle human-driven vehicles. We denote the average number of idle AVs as $N_{i,A}^{I,t}$ and the average number of idle human drivers as $N_{i,H}^{I,t}$. Therefore, the total number of idle vehicles can be expressed as the sum of the two: $N_i^{I,t} = N_{i,A}^{I,t} + N_{i,H}^{I,t}$. In order to describe the relationship between the waiting time $w_{i,t}^p$ and the number of idle vehicles $N_i^{I,t}$, we draw upon the established "square root law" as documented in Li et al. (2021) and Li et al. (2019a), which gives:

$$w_{i,t}^p = \frac{L}{\sqrt{N_i^{I,t}}} = \frac{L}{\sqrt{N_{i,A}^{I,t} + N_{i,H}^{I,t}}}, \qquad i \in \mathcal{N}, t \in \mathcal{T}. \tag{2}$$

where L is a constant model parameter that depends on the size of the zone and the distribution of demand and supply. Eq. (2) indicates that the average passenger waiting time $w_{i,t}^p$ is inversely proportional to the square root value of the number of idle vehicles $N_i^{I,t}$. The rationale behind (2) is straightforward: if all idle vehicles are uniformly distributed over zone i, the distance between two adjacent idle vehicles is inversely proportional to the square root of $N_i^{I,t}$. In the meanwhile, the average distance between two neighboring idle vehicles is proportional to the distance between a passenger and the closest idle vehicle, thus the passenger waiting time $w_{i,t}^p$ satisfies the square root form as deliberated in (2). A rigorous proof of the above intuition can be referred to Proposition 1 of Li et al. (2019a).

[2] Throughout this paper, we refer to mobility-on-demand services as on-demand ride-hailing services where passengers hail a ride in an on-demand manner through a mobile APP, such as the case of Uber and Lyft.

[3] Note that AVs may travel at different speeds compared to human-operated vehicles (either higher or lower), leading to distinct travel times.

[4] This is a reasonable assumption when the zone of the city is large and the vehicle supply is sufficient so that each passenger will be picked up by a nearby vehicle with a high probability. The case of inter-zone matching is left for future studies.

Given that the trip cost for other alternative transport modes is exogenous, the arrival rate of ride-hailing passengers can be determined by the average trip cost c_{ij}^t based on the following demand function:

$$\lambda_{ij}^t = \lambda_{ij}^{0,t} F_p(c_{ij}^t), \qquad i,j \in \mathcal{N}, t \in \mathcal{T}. \tag{3}$$

where $\lambda_{ij}^{0,t}$ denotes the total potential travel demand from zone i to zone j for all transport modes; $F_p(c_{ij}^t)$ represents the percentage of passengers who select ride-hailing as a transport mode. Further, we suppose $F_p(c_{ij}^t)$ is a monotonically decreasing function with respect to c_{ij}^t, indicating that a higher trip cost will result in a lower passenger demand. We emphasize that c_{ij}^t is the average trip cost, while the actual costs for different passengers can be different due to randomness. Eq. (3) implicitly assumes that the arrival rate for ride-hailing passengers depends purely on average trip cost. Through the rest of this paper, we will assume that F_p satisfies the logit model:

$$\lambda_{ij}^t = \lambda_{ij}^{0,t} \frac{e^{-\kappa c_{ij}^t}}{e^{-\kappa c_{ij}^t} + e^{-\kappa c_{ij}^0}}, \qquad i,j \in \mathcal{N}, t \in \mathcal{T}, \tag{4}$$

where κ is the sensitivity parameters and c_{ij}^0 denotes the cost of outside options for trips between zone i and zone j.

Remark 1. We assume that the demand of ride-hailing service is elastic, and that each passenger will decide whether to take ride-hailing services or opt for other alternative modes of transportation (such as buses, railways, and other modes) based on their respective costs. The cost of the ride-hailing platform is endogenous and characterized explicitly in (1). The cost of all other modes is lumped together and treated as exogenous, which is characterized by the cost of the outside option, i.e., c_{ij}^0, in (4). This captures the passenger's choice between ride-hailing and other alternative modes, without the need to explicitly characterize the operational strategies of other modes.

3.2. Incentives of the mixed fleet

We consider the mixed fleet where both AVs and human drivers are deployed to offer on-demand services. We suppose that there is a fixed number of AVs throughout the day, denoted as N_A, which is leased from the automotive original equipment manufacturer (OEM) for mobility-on-demand services. Each AV in the fleet incurs a constant unit cost, including both procurement cost and operational expenditure. All AVs are centrally managed by the platform. Their operational strategies, such as repositioning and spatial distribution of idle vehicles, are directly determined by the platform. In contrast, human drivers make complex decisions regarding whether to join the platform (i.e., choice of jobs), when to join the platform (i.e., choice of work schedules), and where to go for the next ride request (i.e., choice of repositioning locations) in a decentralized manner. The rest of this section characterizes these intricate decisions from three perspectives: (a) choice of jobs; (b) choice of work schedules; and (c) choice of repositioning locations.

Choice of jobs: For-hire human drivers on the TNC platform are typically treated as independent contractors, who choose their jobs based on the prospective earnings. In this case, the total number of driver hours on the platform within the horizon Γ, denoted as N_H, is related to the net earning provided by the ride-hailing platform within the same horizon. Without loss of generality, the total amount of human driver supply function can be modeled as:

$$N_H = N_0 \cdot F_d(\pi). \tag{5}$$

where π denotes the net earning within the horizon paid to the drivers by the platform; N_0 denotes the potential drivers who want to work for the platform; $F_d(\pi)$ is a function that represents the driver's willingness of working for the ride-hailing platform. Naturally, we suppose $F_d(\pi)$ is a monotone-increasing function with respect to π. Consequently, as net earnings increase, we expect a corresponding rise in the number of drivers inclined to participate in the ride-hailing service.

Choice of work schedules: Drivers who have made their job choices will subsequently establish their work schedules. Intuitively, drivers may opt to join the platform during peak periods, when demand and prospective earning are highest, and exit during off-peak periods when demand and prospective earning are lower. However, if a considerable number of drivers join the platform simultaneously, they will compete for passengers, leading to a reduction in average earnings for each driver. This highlights the complex interplay among drivers, ultimately leading to the attainment of an equilibrium in their decisions. Similar to Yang et al. (2005) and Zha et al. (2017), the scheduling decision of the drivers can be characterized using a time-expanded network (as shown in Fig. 1), where nodes correspond to time points within a ride-hailing market, and all drivers start from the same virtual origin, denoted as O, and terminate at a symbolic destination, denoted as D. For each driver, a path between the origin–destination pair defines a work shift, which provides complete information on when the driver enters and exits from the platform during this shift. Let $\mathcal{P}$ denote the set of all possible paths (shifts) in the time-expended graph, and denote n_p as the number of drivers choosing to work in shift $p \in \mathcal{P}$. The total number of human drivers at time t, i.e., N_H^t can be derived as:

$$N_H^t = \sum_{p \in \mathcal{P}} n_p \cdot \delta_p^t, \quad p \in \mathcal{P}, \quad \forall t \in \mathcal{T}. \tag{6}$$

where δ_p^t is equal to 1 if t belongs to path p, and 0 otherwise. Given the supply of human drivers at time t, denoted as N_H^t, we can derive the average earning of human drivers per unit of time as:

$$R_H^t = \frac{(1-\eta) \cdot \sum_{i=1}^N \sum_{j=1}^N \left(r_{ij}^t \cdot \lambda_{ij}^t \cdot d_{ij} \right)}{N_H^t + N_A}, \quad \forall i,j \in \mathcal{N}, t \in \mathcal{T}. \tag{7}$$

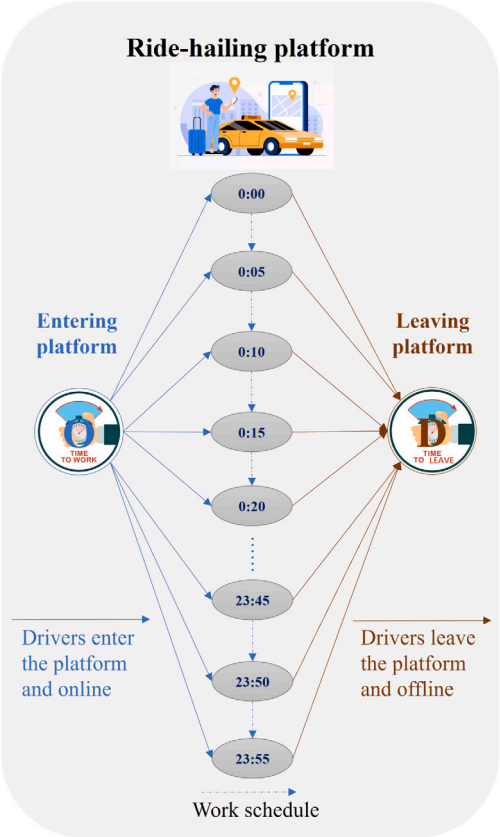


Fig. 1. Time-expanded network for driver working scheduling.

where $\eta \in (0,1)$ represents the commission rate charged by the ride-hailing platform; and the numerator $(1-\eta)\cdot\sum_{i=1}^{N}\sum_{j=1}^{N}\left(r_{ij}^{t}\lambda_{ij}^{t}d_{ij}\right)$ corresponds to the total earning for all drivers that work at time t, including both AVs and human drivers. In (7), if the passenger demand λ_{ij}^{t} and the decisions of the platform (i.e., η, r_{ij}^{t}, N_A) are given, the hourly earning of drivers only depends on the number of human drivers N_H^t: the more human drivers there are, the more competition between human drivers, and the smaller the average driver revenue. With slight abuse of notation, we denote the average earning of human drivers R_H^t as a function of the number of human drivers N_H^t (under the condition that η, r_{ij}^{t}, λ_{ij}^{t} and N_A are given). Using this notation, (7) can be compactly written as:

$$R_H^t = R_H^t(N_H^t), \quad \forall t \in \mathcal{T}. \tag{8}$$

Each human driver chooses his/her work schedule based on the trade-off between total earnings and total cost. In the meanwhile, the decisions of the drivers are intimately correlated due to the competition effect. To capture the equilibrium decisions, we define R_p as the total earning for drivers choosing path (or equivalently, schedule) $p \in \mathcal{P}$. Obviously, R_p can be written as the sum of the earnings of each time period along the path p:

$$R_p = \sum_{t=1}^{\mathcal{T}} R_H^t \cdot \delta_p^t, \quad p \in P. \tag{9}$$

During their scheduled shifts, drivers incur two main types of expenses: (1) a time-based cost, which increases linearly with the length of the working hours t_p along a given path p, including expenses such as fuel, maintenance, and vehicle wear-and-tear; and (2) a disutility that stems from prolonged cumulative working hours t_p, which is presumed to escalate at a rate exceeding linearity in relation to t_p. Specifically, for any chosen path p, the total cost c_p comprises these two components and can be articulated as follows:

$$c_p = \omega_0 \cdot t_p + \omega_1 \cdot (t_p)^{\omega_2}, \quad t^p \in \{\underline{t}^p, \overline{t}^p\}. \tag{10}$$

where ω_0, ω_1 and ω_2 are exogenous model parameters that characterize the cost associated with the cumulative working hours t_p; $\underline{t}^p$ and $\overline{t}^p$ represent the lower bound and upper bound on the accumulated hours of each work shift. Given the earning R_p and cost c_p of each path, drivers will choose their work schedules by comparing the net earning of each path, which can be defined as $\pi_p = R_p - c_p$. We define the equilibrium of driver's scheduling decisions as follows:

Definition 1. At equilibrium, all work schedules $p \in P$ with $n_p > 0$ yield an equal net earning, i.e., earning minus cost $\pi_p = R_p - c_p$, which is no less than that of any p with $n_p = 0$.

Let π^* denote drivers' net earning at equilibrium. We clearly have $\pi_p = \pi^*$ for all the used paths. Throughout the rest of this paper, we define the elastic driver supply model as:

$$N_H = N_0 \cdot F_d(\pi^*) = N_0 \cdot \left(1 - \exp(-\xi \cdot \pi^*)\right), \tag{11}$$

Obviously, the relationship expressed in (11) is invertible, which allows us to derive a function that maps the number of human drivers N_H to the equilibrium net earnings π^*. This inverse relationship is denoted by $\pi^* = F_d^{-1}(N_H)$. The equilibrium state mentioned above is in line with the neoclassical theory of labor supply, as referenced in Farber (2015). Let $\mathbf{n}$ represent the vector of all path flows, i.e., $\mathbf{n} = \{n_p\}_{p \in \mathcal{P}}$. It is a well-established principle that equilibrium decisions within a context of elastic driver supply can be characterized as the solution to the following optimization problem (Yang et al., 2005):

$$\underset{\mathbf{n} \geq 0, N_H \geq 0}{\text{maximize}} \quad \sum_{t \in \mathcal{T}} \left(\int_0^{N_H^t} R_H^t(w) \mathrm{d}w \right) - \sum_{p \in \mathcal{P}} c_p \cdot n_p - \int_0^{N_H} F_d^{-1}(w) \mathrm{d}w \tag{12}$$

$$\begin{cases} \sum_{p \in \mathcal{P}} n_p = N_H, & \text{(a)} \\ \sum_{p \in \mathcal{P}} n_p \cdot \delta_p^t = N_H^t, \quad \forall t \in \mathcal{T}, & \text{(b)} \\ n_p \geq 0, \quad \forall p \in \mathcal{P}. & \text{(c)} \end{cases} \tag{13}$$

where (12) is a virtual objective function that depends on the driver earning R_H^t and the cost c_p; constraint (13a) requires that the total supply of human driver hours is equal to N_H, which is determined by (5); constraint (13b) specifies the relation between path flow and the number of available drivers in each time period. Since $R_H^t(N_H^t)$ is a decreasing function of N_H^t, the optimization problem (12) is clearly convex and thus can be easily solved to obtain its globally optimal solution. This solution corresponds to the equilibrium where each self-interested driver reaches the optimal solution and has no incentives to deviate unilaterally.

Remark 2. We clarify that the "time-expanded network" in our study is designed to characterize the equilibrium of drivers' *scheduling decisions* instead of the routing decisions. In this case, each link in the time-expanded network corresponds to a specific time period of the day (e.g., 10:00 am-10:15 am), instead of a specific road segment over the transportation network, and each work shift is defined as a "path", with the shift's start time as the "origin" and the end time as the "destination". Drivers select from different shifts to maximize their earnings, leading to an equilibrium where no individual can be better off by unilaterally changing her/his choice. Similar to Yang et al. (2005), Zha et al. (2017) and Ke et al. (2019), this model captures the "congestion effect" of drivers' scheduling choices. Specifically, if more drivers select a specific time period in their work schedule, the expected earning for each driver in this time period will decrease due to the increased competition among drivers (see equation (6)). This is similar to "traffic congestion" on the road network in a classic traffic assignment model: when more travelers choose certain link (time period in our case) on the road network, then the traffic congestion on this link is more serious and the travel time (driver earning in our case) will be negatively influenced.

After the solution to (12) is obtained, we can further derive the number of human drivers that enter and exit from the ride-hailing market at any time t, which crucially affects the dynamics of available vehicles on the ride-hailing platform. We denote $\tilde{N}_H^t$ and $\hat{N}_H^t$ as the number of human drivers that enter and exit from the platform at time t, respectively. Let $\underline{\mathbb{1}}_p^t$ be an indicator function, which equals 1 if time t is the first time period within the work schedule p. Similarly, let $\bar{\mathbb{1}}_p^t$ be an indicator function that equals to 1 if time t is the last time period within p. In this case, the number of entering and exiting human drivers, i.e., $\tilde{N}_H^t$ and $\hat{N}_H^t$, depend on $\underline{\mathbb{1}}_p^t$ and $\bar{\mathbb{1}}_p^t$ through the following relations:

$$\begin{cases} \tilde{N}_H^t = \sum_{p \in P} \left(n_p \cdot \underline{\mathbb{1}}_p^t \right), & \text{(a)} \\ \hat{N}_H^t = \sum_{p \in P} \left(n_p \cdot \bar{\mathbb{1}}_p^t \right). & \text{(b)} \end{cases} \tag{14}$$

where $\tilde{N}_H^t$ is the summation of all flows that enter the platform at time t, and $\hat{N}_H^t$ is the summation of all flows that exit from the platform at time t. Note that (14) only characterizes the aggregated number of human drivers without specifying where the driver enters or exists the system. To characterize the spatial distribution of $\tilde{N}_H^t$ and $\hat{N}_H^t$, we further denote $\tilde{N}_{i,H}^t$ and $\hat{N}_{i,H}^t$ as the number of drivers entering and leaving the ride-hailing market through zone i. Since drivers generally enter the market from where they live, and exit randomly according to the demand distribution. Therefore, we assume that the total number of entering drivers $\tilde{N}_H^t$ is split into different zones. This leads to a vector of $\tilde{N}_{i,H}^t$ for each zone according to an exogenous spatial distribution (which may depend on exogenous parameters such as population, potential demand, etc.), whereas the exiting drivers $\hat{N}_H^t$ are split into $\hat{N}_{i,H}^t$ based on the spatial distribution of idle human drivers at time t. In particular, we have:

$$\tilde{N}_{i,H}^t = \frac{W_i}{\sum_{i=1}^N W_i} \cdot \tilde{N}_H^t, \tag{15}$$

$$\hat{N}^t_{i,H} = \frac{N^t_{i,H}}{\sum_{i=1}^N N^t_{i,H}} \cdot \hat{N}^t_H. \tag{16}$$

where W_i is an exogenous parameter corresponding to a proxy for the concentration of human driver's residence in zone i (e.g., population). In this case, a larger value of W_i indicates a higher probability of starting the work shift in zone i.

Choice of repositioning locations: Given the work schedule, each driver will enter the market to provide mobility-on-demand for passengers in complement with a fleet of AVs. However, between rides, idle AVs or idle human drivers both need to decide where to reposition the vehicle, so as to receive the next ride request as soon as possible for a higher earning. As mentioned above, the repositioning strategies of AVs are centrally managed by the platform. However, human drivers selfishly make their own repositioning decisions based on expected earning opportunities in different zones of the city in a decentralized manner. We use $f^t_{ij,A}$ and $f^t_{ij,H}$ to represent the reposition flow from zone i to zone j at time t of AVs and human drivers, respectively. In this case, $f^t_{ij,A}$ is a decision variable for the platform, whereas $f^t_{ij,H}$ is an endogenous variable that depends on the collective decisions of individual drivers. In particular, for each driver, the repositioning decisions depend on the average perceivable earnings in each zone at time t. Consider a driver that successfully takes a ride in zone i at time t. The driver's expected earning e^t_i is a function of the trip duration t_{ij}, ride fare r^t_{ij}, and arrival rate λ^t_{ij}. Specifically,

$$e^t_i = \frac{\sum_{j=1}^N (1-\eta)\left(r^t_{ij} \cdot \lambda^t_{ij} \cdot d_{ij}\right)}{\sum_{j=1}^N \lambda^t_{ij}}, \quad p \in P, \forall t \in \mathcal{T}. \tag{17}$$

We further need to specify the average time consumed by the driver to earn an average payment of e^t_i. For this purpose, we first define $w^d_{i,t}$ as the average waiting time for idle vehicles in zone i at time t to receive the next ride request. Based on Little's law, it satisfies:

$$w^d_{i,t} = \frac{\sum_{j=1}^N \lambda^t_{ij}}{N^{I,t}_i}, \quad p \in P, \forall t \in \mathcal{T}. \tag{18}$$

where the numerator denotes the total passenger flow starting from zone i and the denominator denotes the total number of idle vehicles (including both AV and human drivers) in zone i. Once a driver finishes a trip in zone i, she/he determines whether to stay there or reposition to another zone j to match another passenger. Let t^H_{ij} denote the average trip time for human drivers to travel from zone i to zone j, and denote $\bar{t}^H_i$ as the average trip time for human-driven trips originating from zone i, i.e., $\bar{t}^H_i = \frac{1}{N}\sum_{j=1}^N t^H_{ij}$. Under these notations, we consider two cases: (1) if the driver chooses to stay in zone i, she/he will experience an average waiting time $w^d_{i,t}$ and trip time $\bar{t}^H_i$ before finishing the next trip. The earning utility during this time is $e^t_i/(w^d_{i,t} + \bar{t}^H_i)$; (2) if the driver chooses to reposition to zone j, she/he will undergo waiting time $(w^d_{j,t} + t^H_{ij})$ and the average trip time $\bar{t}_j$. The earning utility in this case is $e^t_j/(w^d_{j,t} + \bar{t}^H_j + \bar{t}^H_j)$. Based on this, we employ a discrete choice model to quantify the probability that the idle human driver in zone i chooses to transfer to zone j, which can be modeled as:

$$\begin{cases} \mathbb{P}^t_{ij} = \dfrac{e^{\sigma e^t_j/(w^t_{d,j}+t^H_{ij}+\bar{t}^H_j)}}{\sum_{k\neq i} e^{\sigma e^t_k/(w^t_{d,k}+t^H_{ik}+\bar{t}^H_k)} + e^{\sigma e^t_i/(w^t_{d,i}+\bar{t}^H_i)}}, & j \neq i, \quad \text{(a)} \\ \mathbb{P}^t_{ii} = \dfrac{e^{\sigma e^t_j/(w^t_{d,j}+\bar{t}^H_j)}}{\sum_{k\neq i} e^{\sigma \bar{e}^t_k/(w^t_{d,k}+t^H_{ik}+\bar{t}^H_k)} + e^{\sigma e^t_i/(w^t_{d,i}+\bar{t}^H_i)}}, & j = i. \quad \text{(b)} \end{cases} \tag{19}$$

where $\mathbb{P}^t_{ij}$ denotes the probability of a driver choosing to transfer to zone i from zone j at time t, whereas $\mathbb{P}^t_{ii}$ represents the driver preferring to stay in zone i to wait for the next ride at time t. The repositioning probability is determined by the expected earnings in each zone i. Based on the repositioning probability, we can model the human driver repositioning flow as:

$$f^t_{ij,H} = \mathbb{P}^t_{ij} \cdot \left(N^{I,t}_{i,H} - \sum_{j=1}^N \left[\frac{N^{I,t}_{i,H}}{N^{I,t}_{i,H} + N^{I,t}_{i,A}} \cdot \lambda^t_{ij} \right] \right), \quad i,j \in \mathcal{N}, t \in \mathcal{T}. \tag{20}$$

The summation term in the right-hand side of (20) can be interpreted as the total trips that start from zone i at time t and are allocated to idle human drivers. Therefore, the difference between $N^{I,t}_{i,H}$ and the summation term represents the rest of the idle human drivers who need to make repositioning decisions in zone i at time t.

3.3. Incentives of the platform

The ride-hailing platform hires a hybrid vehicle fleet (AV and human drivers) to maximize platform profit. The decisions of the platform are made at multiple time resolutions. In the long run (e.g., up to a few months), the platform makes strategic decisions such as the commission rate and the fleet size of AVs, whereas in the short term (e.g., within one day), these long-run decisions are fixed and the platform makes operational decisions over space and time, such as the ride fare r^t_{ij} and the repositioning flow of AV $f^t_{ij,A}$. In this section, we consider long-term decisions (e.g., η and N_A) as exogenous and focus on short-term operational decisions of the platform. In particular, if we denote $\mathbf{r} = \{r^t_{ij}\}_{i\in\mathcal{N},t\in\mathcal{T}}$ and $\mathbf{f}_A = \{f^t_{ij,A}\}_{i,j\in\mathcal{N},t\in\mathcal{T}}$ as the vectors corresponding to ride fare and repositioning flow of AV, respectively. Let $\mathbf{N}^I_A = \{N^{I,t}_{i,A}\}_{i\in\mathcal{N},t\in\mathcal{T}}$ denote the vector for the spatial distribution of idle AVs, and let

$\lambda = \{\lambda_{ij}^t\}_{i,j\in\mathcal{N},t\in\mathcal{T}}$ denote the vector of passenger demand. The profit maximization problem for the ride-hailing platform can be then modeled as follows:

$$\max_{\mathbf{r},\mathbf{f}_A,\mathbf{N}_A^I,\lambda} \quad \sum_{t=1}^{T}\sum_{i=1}^{N}\sum_{j=1}^{N} r_{ij}^t \cdot \lambda_{ij}^t \cdot d_{ij} - C \cdot N_A \tag{21}$$

$$\begin{cases} \lambda_{ij}^t = \lambda_{ij}^{0,t} \cdot F_p(\alpha \cdot w_{i,t}^p + \beta \cdot t_{ij}^t + r_{ij}^t d_{ij}), \qquad i,j \in \mathcal{N}, t \in \mathcal{T}, & \text{(a)} \\ w_{i,t}^p = \dfrac{L}{\sqrt{N_{i,A}^{I,t}+N_{i,H}^{I,t}}} \le w_{max}^p, \quad r_{ij}^t \le \bar{r}, \qquad i,j \in \mathcal{N}, t \in \mathcal{T}, & \text{(b)} \\ N_{i,A}^{I,t} = N_{i,A}^0 - \sum\limits_{\tau=1}^{t}\sum\limits_{j=1}^{N}\left(\dfrac{N_{i,A}^{I,\tau}}{N_{i,A}^{I,\tau}+N_{i,H}^{I,\tau}} \cdot \lambda_{ij}^\tau + f_{ij,A}^\tau\right) & \text{(c)} \\ \qquad + \sum\limits_{\tau=1}^{t}\sum\limits_{j\in\{j|\Delta T\cdot(t-1)\le\Delta T\cdot(\tau-1)+t_{ij}^A\le\Delta T\cdot t\}}\left(\dfrac{N_{j,A}^{I,\tau}}{N_{j,A}^{I,\tau}+N_{j,H}^{I,\tau}} \cdot \lambda_{ji}^\tau + f_{ji,A}^\tau\right), & \\ N_{i,H}^{I,t} = N_{i,H}^0 - \sum\limits_{\tau=1}^{t}\sum\limits_{j=1}^{N}\left(\dfrac{N_{i,H}^{I,\tau}}{N_{i,H}^{I,\tau}+N_{i,A}^{I,\tau}} \cdot \lambda_{ij}^\tau + f_{ij,H}^\tau\right) - \sum\limits_{\tau=1}^{t}\hat{N}_{i,H}^\tau + \sum\limits_{\tau=1}^{t}\tilde{N}_{i,H}^\tau & \text{(d)} \\ \qquad + \sum\limits_{\tau=1}^{t}\sum\limits_{j\in\{j|\Delta T\cdot(t-1)\le\Delta T\cdot(\tau-1)+t_{ij}^H\le\Delta T\cdot t\}}\left(\dfrac{N_{j,H}^{I,\tau}}{N_{j,A}^{I,\tau}+N_{j,H}^{I,\tau}} \cdot \lambda_{ji}^\tau + f_{ji,H}^\tau\right), & \\ f_{ij,H}^t = \mathbb{P}_{ij}^t \cdot \left(N_{H,i}^{I,t} - \sum\limits_{j=1}^{N}\left[\dfrac{N_{i,H}^{I,t}}{N_{i,H}^{I,t}+N_{i,A}^{I,t}} \cdot \lambda_{ij}^t\right]\right), \qquad i,j \in \mathcal{N}, t \in \mathcal{T}, & \text{(e)} \\ N_{i,A}^{I,t} \ge 0, N_{i,H}^{I,t} \ge 0, \quad i,j \in \mathcal{N}, t \in \mathcal{T}. & \text{(f)} \end{cases} \tag{22}$$

where C is the cost of AV per unit time, t_{ij}^A is the average time for AVs to travel from zone i to zone j. Typically the speed of AVs can be different from that of HVs, thus t_{ij}^A is not necessarily equal to t_{ij}^H. In this paper, we assume that t_{ij}^A can be either larger or smaller than t_{ij}^H, accommodating both cases. Furthermore, t_{ij}^A, t_{ij}^H, and t_{ij}^t are related through the following equation:

$$t_{ij}^t = \frac{N_{i,H}^{I,t}}{N_{i,H}^{I,t}+N_{i,A}^{I,t}} t_{ij}^H + \frac{N_{i,A}^{I,t}}{N_{i,H}^{I,t}+N_{i,A}^{I,t}} t_{ij}^A, \tag{23}$$

where the ride-hand side of (23) is derived based on the premise that passengers are dispatched to the closest idle vehicle, regardless of whether it is an AV or a human driver.[5] If we further assume that all vehicles are uniformly distributed within each zone, then the probability of a passenger in zone i being assigned to an idle human driver at time t is $\frac{N_{i,H}^{I,t}}{N_{i,H}^{I,t}+N_{i,A}^{I,t}}$, whereas the probability of a passenger in zone i being assigned to an idle AV at time t is $\frac{N_{i,A}^{I,t}}{N_{i,H}^{I,t}+N_{i,A}^{I,t}}$. The objective function (21) defines the overall platform profit. The first term in (21) represents the total revenue from passengers (minus the payment to human drivers), and the second term in (21) denotes the cost of deploying the AV fleet. Since we consider short-term operational decisions and treat η and N_A as fixed, (21) is equivalent to maximizing the total revenue only. Note that we include the endogenous variable λ as part of the decision variable. This is mainly for the convenience of developing solution algorithms (to be discussed later), and it does not change the solution of the optimization problem as far as (22a) is satisfied.

In the above profit-maximization problem, constraint (22a) corresponds to the demand of ride-hailing passengers. Constraint (22b) requires the pickup time in each zone and per-time ride fare between each pair of orientation-destination (OD) are both smaller than an upper bound. Constraints (22c) and (22d) record the number of idle AV and human drivers in each zone at each time. Specifically, in Eq. (22c), $N_{i,A}^0$ represents the initial deployment of AVs in zone i at time 0. At each time, the number of available AVs in zone i is updated, taking into account passenger demand and the flow of repositioned AVs to and from station i in the time interval from time 0 to time t. In particular, the second term in the right-hand side of (22c) denotes the number of AVs (including both occupied AVs and repositioning AVs) that exit zone i at time t, and third term in the right-hand side of (22c) denotes the AVs that were dispatched to travel to zone i at a previous time (e.g., $\tau \le t$), but arrives in zone i at time t (i.e., between $\Delta T \cdot (t-1)$ and $\Delta T \cdot t$). Similarly, in Eq. (22d), $N_{i,H}^0$ is the initial deployment of human drivers in zone i, and $\tilde{N}_H^t$ and $\hat{N}_H^t$ denote the number of human drivers entering and leaving the ride-hailing platform in zone i at time t, respectively. These variables depend on the working choices and schedules of human drivers, which has been modeled in the previous subsection. Note that we can rewrite

[5] Note that this dispatching policy is the most efficient because, compared to assigning passengers to a specific type of vehicle, it can significantly reduce waiting times.

(22c) in an iterative form as follows:

$$N_{i,A}^{I,t+1} = N_{i,A}^{I,t} - \sum_{j=1}^{N} \left(\frac{N_{i,A}^{I,t}}{N_{i,H}^{I,t} + N_{i,A}^{I,t}} \cdot \lambda_{ij}^{t} + f_{ij,A}^{t} \right) + \sum_{\tau=1}^{t} \sum_{j \in \{j \mid \Delta T \cdot (t-1) \le \Delta T \cdot (\tau-1) + t_{ij}^{A} \le \Delta T \cdot t\}} \left(\frac{N_{j,A}^{I,\tau}}{N_{j,A}^{I,\tau} + N_{j,H}^{I,\tau}} \cdot \lambda_{ji}^{\tau} + f_{ji,A}^{\tau} \right). \tag{24}$$

Similarly, (22d) can be also reformulated as:

$$N_{i,H}^{I,t+1} = N_{i,H}^{I,t} - \sum_{j=1}^{N} \left(\frac{N_{i,H}^{I,t}}{N_{i,H}^{I,t} + N_{i,A}^{I,t}} \cdot \lambda_{ij}^{t} + f_{ij,H}^{t} \right) - \hat{N}_{i,H}^{t} + \tilde{N}_{i,H}^{t} + \sum_{\tau=1}^{t} \sum_{j \in \{j \mid \Delta T \cdot (t-1) \le \Delta T \cdot (\tau-1) + t_{ij}^{H} \le \Delta T \cdot t\}} \left(\frac{N_{j,H}^{I,\tau}}{N_{j,A}^{I,\tau} + N_{j,H}^{I,\tau}} \cdot \lambda_{ji}^{\tau} + f_{ji,H}^{\tau} \right). \tag{25}$$

Clearly, (24) and (25) are equivalent to (22c) and (22d), respectively.

Remark 3. The platform's profit maximization problem (21) is a multi-stage optimization spanning an extensive time horizon (such as an entire day), which necessitates prior knowledge of all future demand information. However, it turns out that when human drivers' decisions are predetermined, the dynamics of the ride-hailing market are short-term in memory: decisions made in the morning are unlikely to significantly affect afternoon demand patterns. Under these circumstances, problem (21) can be resolved with sufficient precision by addressing a truncated horizon at each time step, which only requires demand forecasting for the immediate future (e.g., the forthcoming 45 min). The specifics of this method will be elaborated in subsequent sections of the paper.

Remark 4. We clarify that in essence, whether or not traffic congestion is accounted for does not markedly alter the system dynamics that we aim to capture. Specifically, traffic congestion mainly impacts the travel time between different zones, which further affects the arrival patterns of vehicles. For example, increased congestion would lead to longer travel times to destinations, resulting in a delayed vehicle arrival in the destination zone after it is sent out. This will mainly influence the third term on the right-hand side of Eqs. (24) and (25): if traffic is more congested, then vehicles arrive later, and the correspond third term will show up in later time steps. However, even if congestion levels worsen or improve, and thereby might affect arrival times, such impacts are minor relative to the time granularity of our model (i.e., 5 min), and thus do not significantly affect the dynamics described by Eqs. (24) and (25). For instance, empirical studies have shown that ride-hailing vehicles contribute to about 50% of the traffic congestion in San Francisco, which increases by 26%–27% with their presence in the market (Castiglione et al., 2018). This suggests that the addition of several thousand ride-hailing vehicles can altogether reduce traffic speed by approximately 13–13.5% (i.e., half of 26%–27%). In the context of our study, the average ride-hailing trip in San Francisco lasts about 12 min. If a change in the number of ride-hailing vehicles alters traffic speed by 10% (note that this is already a very safe estimation, considering the complete removal of all these vehicles would only affect speeds by 13%), the average trip duration would be between 10.8 to 13.2 min. Within our discrete-time model at 5-minute granularity, all vehicles would become idle in destination zones after three time intervals, as shown in Eqs. (24) and (25), regardless of congestion levels of the transportation network. Consequently, the congestion caused by ride-hailing services does not significantly impact the idle vehicle dynamics, supporting the use of static travel times for each origin–destination pair in our analysis.

3.4. Market equilibrium

In the ride-hailing market, human drivers and the ride-hailing platform make decisions to maximize their own benefits while interacting with each other. The outcome of this interaction is defined as the market equilibrium, which can be characterized through a game-theoretic framework. In particular, when the work schedule of human drivers is fixed, the total number of human drivers that enter or exit from the ride-hailing platform at each time, i.e., $\tilde{N}_{i,H}^{t}$ and $\hat{N}_{i,H}^{t}$, is also fixed correspondingly. In this case, the ride-hailing platform determines the optimal ride fares $\mathbf{r}$ and repositioning flows $\mathbf{f}_A$ under fixed $\tilde{N}_{i,H}^{t}$ and $\hat{N}_{i,H}^{t}$ to maximize the profit, as characterized by (21). On the other hand, when the operational decisions of the platform are fixed, the prospective earning of drivers in each time period can be also calculated. In this case, drivers will make scheduling decisions accordingly to maximize individual net earnings, the equilibrium of which can be characterized by (12). Overall, we consider a problem comprised of two interdependent optimization: one corresponding to the decisions of the platform, and the other one corresponding to the decisions of the virtual player that represents all the human drivers collectively. These two optimization problems are related as the decisions of the platform and the decisions of the virtual player can influence the objective value of each other. Therefore, the market equilibrium can be characterized as the Nash equilibrium of a two-player game, where player 1 is the ride-hailing platform that solves (21) when the work schedule of drivers (i.e., $\tilde{N}_{i,H}^{t}$ and $\hat{N}_{i,H}^{t}$) are given, and player 2 is a virtual player that represents all the human drivers to collectively solve (12) when the operational decision of the ride-hailing platform (i.e., $\mathbf{r}$, $\mathbf{f}_A$, $\mathbf{N}_A^I$, λ) are

given. At the Nash equilibrium, both players are satisfied given that the other player also players the Nash equilibrium, thus the system interaction reaches a stable outcome.

To be more specific, we define $\mathbf{n}^*$ as the optimal solution to (12), and denote $\tilde{N}^{t*}_{i,H}$ and $\hat{N}^{t*}_{i,H}$ as the corresponding number of entering and exiting drivers, which relates to $\mathbf{n}^*$ through (14) and (15). Clearly, since $\mathbf{n}^*$ depend on the decisions of the ride-hailing platform (i.e., $\mathbf{r}$, $\mathbf{f}_A$, $\mathbf{N}^I_A$, λ), with slight abuse of notation, $\mathbf{n}^*$ can be written as a function, i.e., $\mathbf{n}^* = \mathbf{n}(\mathbf{r}, \mathbf{f}_A, \mathbf{N}^I_A, \lambda)$. Similarly, denote the optimal solution to the ride-hailing platform's profit maximization problem (21) as $(\mathbf{r}^*, \mathbf{f}^*_A, \mathbf{N}^{I*}_A, \lambda^*)$, then all these decisions are functions of $\tilde{N}^t_{i,H}$ and $\hat{N}^t_{i,H}$. With slight abuse of notation, we define them in functional forms, i.e., $\mathbf{r}^* = \mathbf{r}(\tilde{N}^t_{i,H}, \hat{N}^t_{i,H}), \mathbf{f}^*_A = \mathbf{f}_A(\tilde{N}^t_{i,H}, \hat{N}^t_{i,H}), \mathbf{N}^{I*}_A = \mathbf{N}^I_A(\tilde{N}^t_{i,H}, \hat{N}^t_{i,H}), \lambda^* = \lambda(\tilde{N}^t_{i,H}, \hat{N}^t_{i,H})$ Using the above notation, the market equilibrium is defined as the Nash equilibrium for two-player game, i.e., $(\mathbf{n}^*, \tilde{N}^{t*}_{i,H}, \hat{N}^{t*}_{i,H}, \mathbf{r}^*, \mathbf{f}^*_A, \mathbf{N}^{I*}_A, \lambda^*)$, which satisfies:

$$\begin{cases} \mathbf{n}^* = \mathbf{n}(\mathbf{r}^*, \mathbf{f}^*_A, \mathbf{N}^{I*}_A, \lambda^*), & \text{(a)} \\ \mathbf{r}^* = \mathbf{r}(\tilde{N}^{t*}_{i,H}, \hat{N}^{t*}_{i,H}), & \text{(b)} \\ \mathbf{f}^*_A = \mathbf{f}_A(\tilde{N}^{t*}_{i,H}, \hat{N}^{t*}_{i,H}), & \text{(c)} \\ \mathbf{N}^{I*}_A = \mathbf{N}^I_A(\tilde{N}^{t*}_{i,H}, \hat{N}^{t*}_{i,H}), & \text{(d)} \\ \lambda^* = \lambda(\tilde{N}^{t*}_{i,H}, \hat{N}^{t*}_{i,H}), & \text{(e)} \end{cases} \tag{26}$$

where Eq. (26a) indicates that the decision of the drivers is the best response to the platform's decision, and Eqs. (26b)–(26e) indicate that the platform's decision is the best response to the drivers' scheduling decisions. Note that $\tilde{N}^{t*}_{i,H}$ and $\hat{N}^{t*}_{i,H}$ depend on $\mathbf{n}^*$ through (14).

Remark 5. The Nash equilibrium defined in (26) is a vector encompassing the platform's decisions (such as ride fare, repositioning flow, and spatial distribution of idle AVs) and the collective scheduling choices of human drivers (such as the number of drivers on each shift, the entry and exit of driver in each hour, etc.). Within the Nash equilibrium, the outcomes pertaining to the platform directly reflect the decisions made by the platform, while the outcomes pertaining to human drivers represent the cumulative effect of their collective choices, instead of the individual decisions of each driver. This collective modeling of human driver behavior is sufficient for our analysis because knowing the overall impact of human drivers' decisions is adequate for the platform to form its strategies and for us to derive the overall market equilibrium.

Remark 6. The proposed game-theoretic framework assumes that both the platform and drivers are self-interested, and importantly, the human drivers' scheduling decisions cannot be dictated by the platform. We maintain that this is a reasonable assumption because, in practice, the process of drivers entering and exiting ride-hailing platforms, such as Uber, Lyft, and Didi, is primarily determined by the drivers themselves in response to day-to-day fluctuations in earning opportunities. This is precisely the case captured by the game-theoretic model proposed in our paper.

Remark 7. The proposed game-theoretical model involves human drivers' long-term, strategic scheduling decisions and the platform's real-time, operational decisions. Although the time scales of the decisions differ, these decision-making processes are reconciled in the sense that they are made to optimize the long-term objectives of the platform and human drivers over a consistent time horizon, i.e., a day. Due to the involvement of long-term decisions, the game may take a considerable time to reach an equilibrium state, exhibiting non-equilibrium states in the interim. While exploring the properties of these non-equilibrium states is an interesting direction, we leave this as future work.

4. Solution method

The formulated two-player game can be solved by a standard best response algorithm, where each player iteratively computes the optimal decisions in response to the given decisions of the other player until it converges. However, since the platform's profit maximization problem (21) is non-convex, even if a candidate Nash equilibrium can be derived from the best response algorithm, it is still difficult to validate whether the candidate solution is indeed a Nash equilibrium. This is because for a candidate Nash equilibrium $(\mathbf{n}^*, \tilde{N}^{t*}_{i,H}, \hat{N}^{t*}_{i,H}, \mathbf{r}^*, \mathbf{f}^*_A, \mathbf{N}^{I*}_A, \lambda^*)$, although we are confident that $\mathbf{n}^*$ is the globally optimal solution to (12), the operational decisions of the platform (i.e., $\mathbf{r}^*$, $\mathbf{f}^*_A$, $\mathbf{N}^{I*}_A$, λ^*) can be a local solution to the non-convex problem (21). To tackle this challenge, we consider a convex relaxation for (21), which offers an upper bound on the distance between the objective value at the candidate solution and that at the globally optimal solution to (21). This establishes the candidate solution derived from the best response algorithm as an ϵ-Nash equilibrium (Li et al., 2019b). Note that ϵ-Nash equilibrium is an approximation of the exact Nash equilibrium, and the value of ϵ characterizes the approximation error. In particular, a smaller value of ϵ indicates a smaller approximation error, and in the case of $\epsilon = 0$, the approximation becomes exact and ϵ-Nash equilibrium becomes identical to the exact Nash equilibrium. Below we describe the procedure of convex relaxation and the derivation of the approximate Nash equilibrium.

4.1. Convex relaxation

To derive a convex relaxation for (21), we first combine (24) and (25) to obtain its summation as:

$$N_i^{I,t+1} = N_i^{I,t} - \sum_{j=1}^{N}\left(\lambda_{ij}^t + f_{ij}^t\right) + \sum_{\tau=1}^{t}\sum_{j\in\{j|\Delta T\cdot(t-1)\le\Delta T\cdot(\tau-1)+t_{ij}^H\le\Delta T\cdot t\}}^{N}\left(\lambda_{ji}^\tau + f_{ji}^\tau\right) - \hat{N}_{i,H}^t + \tilde{N}_{i,H}^t. \tag{27}$$

where $N_i^{I,t}$ and f_{ij}^t represent the total number of idle vehicles and repositioning flows at time i, respectively, including both AVs and human drivers. Therefore, we have $N_i^{I,t} = N_{i,A}^{I,t} + N_{i,H}^{I,t}$, and $f_{ij}^t = f_{ij,A}^t + f_{ij,H}^t$. Note that in the original formulation, the platform directly determines the repositioning flow of AVs, while the repositioning flow of human drivers is determined through (22e). Intuitively, we can apply a change of variable and regard f_{ij}^t (instead of $f_{ij,A}^t$) as the decision variable without affecting the solution to the optimization problem. This is because if we do not consider the positivity constraints (22f), then the platform can always realize a desired f_{ij}^t by setting $f_{ij,A}^t$ corresponding to a given human drivers' repositioning decisions $f_{ij,H}^t$. Following the above intuition, we consider the following revised profit maximization problem for the platform:

$$\max_{\mathbf{r},\mathbf{f},\mathbf{N}^I,\lambda} \quad \sum_{t=1}^{T}\sum_{i=1}^{N}\sum_{j=1}^{N} r_{ij}^t\cdot\lambda_{ij}^t\cdot d_{ij} - C\cdot N_A \tag{28}$$

$$\begin{cases} \lambda_{ij}^t = \lambda_{ij}^{0,t}\cdot F_p(\alpha\cdot w_{i,t}^p + \beta\cdot\tilde{t}_{ij} + r_{i,j}^t\cdot d_{ij}), \qquad i,j\in\mathcal{N}, t\in\mathcal{T}, & \text{(a)}\\ w_{i,t}^p = \dfrac{L}{\sqrt{N_i^{I,t}}} \le w_{max}^p, \quad r_{ij}^t \le \bar{r}_{ij}, \qquad i,j\in\mathcal{N}, t\in\mathcal{T}, & \text{(b)}\\ N_i^{I,t+1} = N_i^{I,t} - \sum_{j=1}^{N}\left(\lambda_{ij}^t + f_{ij}^t\right) + \sum_{\tau=1}^{t}\sum_{j\in\{j|\Delta\cdot(t-1)\le\Delta\cdot(\tau-1)+t_{ij}^H\le\Delta\cdot t\}}^{N}\left(\lambda_{ji}^\tau + f_{ji}^\tau\right) & \\ \qquad -\hat{N}_{i,H}^t + \tilde{N}_{i,H}^t, \qquad i,j\in\mathcal{N}, t\in\mathcal{T}, & \text{(c)}\\ N_i^{I,t} \ge 0, \qquad i\in\mathcal{N},\ t\in\mathcal{T}. & \text{(d)} \end{cases} \tag{29}$$

where $\tilde{t}_{ij} = \min\{t_{ij}^H, t_{ij}^A\}$, $\mathbf{f} = \{f_{ij}^t\}_{i,j\in\mathcal{N},t\in\mathcal{T}}$ denotes the total repositioning flow including both AVs and human drivers, and $\mathbf{N}^I = \{N_i^{I,t}\}_{i\in\mathcal{N},t\in\mathcal{T}}$ denotes the vector of total number of idle drivers at each time t in each zone i. Note that the above model (28) can be regarded as a relaxation for the original problem (21). In particular, the average trip time t_{ij}^t is reduced to $\tilde{t}_{ij} = \min\{t_{ij}^H, t_{ij}^A\}$, which implicitly assumes AVs and human drivers both travel at the speed of whichever is faster, leading to an optimistic estimation that represents a relaxation of the original problem. In addition, by comparing (28) and (21), it is clear that (28) relaxes the positivity constraints on $N_{i,A}^{I,t}$ and $N_{i,H}^{I,t}$, i.e., constraint (22f), and replaces it with another constraint which only requires the summation of $N_{i,A}^{I,t}$ and $N_{i,H}^{I,t}$ to be positive, i.e., (29d). Because (22f) is relaxed, it is no longer necessary to track the number of idle vehicles for AVs and human drivers separately, thus we can replace (22d) and (22e) with (29c). Intuitively, the relaxed problem can be viewed as an ideal case where both AVs and human drivers can drive as fast as human drivers, and the platform can achieve any total rebalancing flows by adjusting the repositioning of idle AVs,[6] while satisfying (29d). Therefore, (28) provides an upper bound for the original problem (21). More importantly, we can show that (28) is convex under certain conditions:

Proposition 1. *If for each $i\in\mathcal{N}$, there exists an X_i that satisfies the following two relations:*

$$\begin{cases} \frac{3}{\kappa}\sqrt{X_i} = \alpha\cdot L\cdot\max_{j\in\mathcal{N}}\left[1 - F_p\left(\dfrac{\alpha L}{\sqrt{X_i}} + \beta\tilde{t}_{ij} + \bar{r}_{ij}d_{ij}\right)\right]^2, & \textit{(a)}\\ X_i \le \left(\dfrac{L}{w_{max}^p}\right)^2. & \textit{(b)} \end{cases} \tag{30}$$

then the relaxed profit-optimization problem (21) is concave.

Proof. We first derive the conditions under which the relaxed profit maximization problem (21) is convex, then we show that these convexity conditions are satisfied if (30) is valid. In particular, we first observe that (21) can be viewed as an optimization over the decision variables $\mathbf{f}, \mathbf{N}^I$ and λ, because $\mathbf{r}$ can be derived if other variables are known. For instance, we can plug (29b) into (29a), and calculate the inverse function of $F_p(\cdot)$. Based on the logit model (4), r_{ij}^t should satisfy:

$$r_{ij}^t = \frac{1}{d_{ij}}\left(\frac{1}{\kappa}\cdot\ln\left(\frac{\lambda_{ij}^{0,t} - \lambda_{ij}^t}{\lambda_{ij}^t}\right) + c_{ij}^0 - \beta\cdot\tilde{t}_{ij} - \alpha\cdot\frac{L}{\sqrt{N_i^{I,t}}}\right). \tag{31}$$

[6] Note that the repositioning cost of AVs is already included in the per-time operational cost C, which is independent of the vehicle status. This is a reasonable assumption if AVs travel at the same speed in distinct statuses (e.g., cruising for passengers, repositioning to a distinct zone, carry a passenger from one zone to another).

Define $P_{ij}^t = r_{ij}^t \cdot \lambda_{ij}^t \cdot d_{ij}$ as the platform revenue corresponding to trips from zone i to j at time t, then we have

$$\begin{aligned} P_{ij}^t &= r_{ij}^t \cdot \lambda_{ij}^t \cdot d_{ij} \\ &= \left[\frac{1}{d_{ij}}\left(\frac{1}{\kappa}\cdot \ln\left(\frac{\lambda_{ij}^{0,t}-\lambda_{ij}^t}{\lambda_{ij}^t}\right) + c_{ij}^0 - \beta\cdot\tilde{t}_{ij} - \alpha\cdot\frac{L}{\sqrt{N_i^{I,t}}}\right)\right]\cdot\lambda_{ij}^t\cdot d_{ij} \\ &= \left[\frac{1}{\kappa}\cdot \ln\left(\frac{\lambda_{ij}^{0}-\lambda_{ij}^t}{\lambda_{ij}^t}\right) + c_{ij}^{0,t} - \beta\cdot\tilde{t}_{ij} - \alpha\cdot\frac{L}{\sqrt{N_i^{I,t}}}\right]\cdot\lambda_{ij}^t. \end{aligned} \tag{32}$$

Since the constraints (29c) and (29d) are linear, the relaxed profit-maximization problem (28) is convex as long as P_{ij}^t is convex. Obviously, the P_{ij}^t is a nonlinear function over two variables, i.e., λ_{ij}^t and N_i^t. To check the concavity of $P_{ij}^t(\lambda_{ij}^t, N_i^{I,t})$, we compute the Hessian matrix of P_{ij}^t as:

$$\mathbf{H}_{\mathbf{P}_{\mathbf{ij}}^{\mathbf{t}}} = \begin{bmatrix} \frac{\partial^2 P_{ij}^t}{\partial {N_i^{I,t}}^2} & \frac{\partial^2 P_{ij}^t}{\partial N_i^{I,t}\partial\lambda_{ij}^t} \\ \frac{\partial^2 P_{ij}^t}{\partial\lambda_{ij}^t\partial N_i^{I,t}} & \frac{\partial^2 P_{ij}^t}{\partial {\lambda_{ij}^t}^2} \end{bmatrix} = \begin{bmatrix} -\frac{3}{4}\cdot\alpha\cdot\lambda_{ij}^t\cdot\frac{L}{\left(N_i^{I,t}\right)^{\frac{5}{2}}} & \frac{1}{2}\cdot\alpha\cdot\frac{L}{\left(N_i^{I,t}\right)^{\frac{3}{2}}} \\ \frac{1}{2}\cdot\alpha\cdot\frac{L}{\left(N_i^{I,t}\right)^{\frac{3}{2}}} & -\frac{1}{\kappa\cdot\lambda_{ij}^t}\cdot\left(\frac{\lambda_{ij}^{0,t}}{\lambda_{ij}^{0,t}-\lambda_{ij}^t}\right)^2 \end{bmatrix}. \tag{33}$$

The necessary and sufficient condition for Hessian matrix $\mathbf{H}_{\mathbf{P}_{\mathbf{ij}}^{\mathbf{t}}}$ to be negative definite is

$$\frac{3}{\kappa}\cdot\left(\frac{\lambda_{ij}^{0,t}}{\lambda_{ij}^{0,t}-\lambda_{ij}^t}\right)^2 > \frac{\alpha\cdot L}{\sqrt{N_i^{I,t}}}, \quad i,j\in\mathcal{N},\ t\in\mathcal{T}. \tag{34}$$

Next, we show that if for each $i\in\mathcal{N}$, there exists an X_i that satisfies (30), then (34) is always valid for any decision variables in the feasible regime, i.e., satisfy (29). First, because $F_p(c_{ij}^t)$ is a decreasing function of $F_p(c_{ij}^t)$, then the left hand side of (30a) is strictly increasing with respect to X_i, and the right-hand side of (30a) is decreasing with respect to X_i. As the left-hand side is smaller than the right-hand side when $X_i = 0$, this indicates that there exists a unique X_i that satisfies (30a). For any feasible $N_i^{I,t}$ that satisfies (29b), we have:

$$N_i^{I,t} \geq \left(\frac{L}{w_{max}^p}\right)^2, \quad i\in\mathcal{N},\ t\in\mathcal{T}. \tag{35}$$

Combining with (30b), we can conclude that $N_i^{I,t} \geq X_i$. Furthermore, based on the monotonicity of the left-hand side and right-hand side of (30a), if we replace X_i with the feasible $N_i^{I,t}$ in (35), then we have:

$$\frac{3}{\kappa}\sqrt{N_i^{I,t}} \geq \alpha\cdot L\cdot\max_{j\in\mathcal{N}}\left[1-F_p\left(\frac{\alpha L}{\sqrt{N_i^{I,t}}}+\beta\tilde{t}_{ij}+\bar{r}_{ij}d_{ij}\right)\right]^2, \quad i,j\in\mathcal{N},\ t\in\mathcal{T}. \tag{36}$$

Since the left-hand side of (36) is larger than the max of the right-hand side of all $j\in\mathcal{N}$, then (36) is equivalent to:

$$\frac{3}{\kappa}\sqrt{N_i^{I,t}} \geq \alpha\cdot L\cdot\left[1-F_p\left(\frac{\alpha L}{\sqrt{N_i^{I,t}}}+\beta\tilde{t}_{ij}+\bar{r}_{ij}d_{ij}\right)\right]^2, \quad i,j\in\mathcal{N}, t\in\mathcal{T}. \tag{37}$$

Furthermore, if r_{ij}^t is feasible, then based on (29b), we have $r_{ij}^t \leq \bar{r}_{ij}$, which indicates that:

$$\frac{3}{\kappa}\sqrt{N_i^{I,t}} \geq \alpha\cdot L\cdot\left[1-F_p\left(\frac{\alpha L}{\sqrt{N_i^{I,t}}}+\beta\tilde{t}_{ij}+r_{ij}^t d_{ij}\right)\right]^2 = \alpha\cdot L\cdot\left(1-\frac{\lambda_{ij}^t}{\lambda_{ij}^{0,t}}\right)^2, \quad i,j\in\mathcal{N}, t\in\mathcal{T}. \tag{38}$$

This is clearly equivalent to (34), which completes the proof. □

Proposition 1 provides an important set of conditions, i.e., (30), under which the concavity of the relaxed problem (28) can be established. It is very simple to validate (30): given the exogenous model parameters, we can easily derive the unique solution X_i to (30a), and then validate whether the solution satisfies the inequality in (30b). We would like to emphasize that these concavity conditions (30) are quite mild conditions, which holds under most reasonable model parameters. Our numerical test indicates that these conditions are mainly to avoid the case of "wild goose chase" (Castillo et al., 2022), where there is a very small number of idle vehicles available, and each passenger has to endure an excessively long waiting time, leading to the violation of the concavity condition. In practice, the platform would prefer to avoid entering this regime by setting a reasonable value of upper bound on waiting time. We will validate the above observations in the numerical simulation.

4.2. Approximate Nash equilibrium

In this subsection, we exploit the concavity of the relaxed problem (28) to derive an upper bound for the optimal value of the original problem (21). This allows us to establish the candidate solution obtained from the best response algorithm as an approximate Nash equilibrium, specifically the ϵ-Nash equilibrium (Li et al., 2019b). Moreover, the upper bound we obtain can be used to evaluate the quality of this approximation by providing an estimation of the value of ϵ. The technical details of this procedure will be described below.

As previously discussed, we will first use the best response algorithm to compute a candidate solution for the two-player game. In particular, the two players will alternatively solve the best response problems using the standard interior point method while considering the decision of the competing player as given. For instance, the platform determines the operational decision (i.e., $\mathbf{r}$, $\mathbf{f}_A$, $\mathbf{N}_A^I$, λ) by solving (21) under a given value of $\tilde{N}_{i,H}^t$ and $\tilde{N}_{i,H}^t$, while the other player determines the scheduling decisions of the human drivers (i.e., $\mathbf{n}$) by solving (12) under a given value of platform decisions (i.e., $\mathbf{r}$, $\mathbf{f}_A$, $\mathbf{N}_A^I$, λ), and in the meanwhile, $\tilde{N}_{i,H}^t$ and $\tilde{N}_{i,H}^t$ relate to $\mathbf{n}$ through (14) and (15). Suppose that this iterative process converges, then we can obtain a candidate Nash equilibrium, denoted as $(\mathbf{n}^*, \tilde{N}_{i,H}^{t*}, \hat{N}_{i,H}^{t*}, \mathbf{r}^*, \mathbf{f}_A^*, \mathbf{N}_A^{I*}, \lambda^*)$. However, it is imperative to recognize that the platform's profit maximization problem (21) is non-concave, attributed to nonlinear equality constraints and non-concave objective functions relative to the decision variables. In this case, it is well-known that the interior-point method used in each step of the best response algorithm might not provide a globally optimal solution to (21): the most we can hope for is convergence to a locally optimal solution. For this reason, it is difficult to validate whether the candidate solution is indeed a Nash equilibrium that satisfies (26a), because $\mathbf{r}^*, \mathbf{f}_A^*, \mathbf{N}_A^{I*}$ and λ^* may not be the globally optimal solution to (21). If they are not, then adopting the equilibrium strategy may not coincide with the platform's best interest. However, we note that for the derived candidate Nash equilibrium, even if it is a local solution, it could still be a good approximation of the Nash equilibrium in the original game, if the gap between the derived objective value of (21) and the globally optimal value of (28) is not significant and can be numerically characterized.

To characterize the approximate Nash equilibrium, we note that in practice, each player in the game may be indifferent to a small change in its objective function, as the change may only account for a negligible portion of the objective value. This observation motivates a more relaxed solution concept, which encompasses a broader range of equilibrium solutions where players will not unilaterally change their strategies as long as the current solution is approximately optimal. More formally, we can define such a relaxed solution as the ϵ-Nash equilibrium (Daskalakis et al., 2006; Li et al., 2019b). For notation convenience, let us define $\pi_1(\mathbf{n}, \mathbf{r}, \mathbf{f}_A, \mathbf{N}_A^I, \lambda)$ as the objective value of the platform, which depends on the decisions of both players and corresponds to (21). Denote $\pi_2(\mathbf{n}, \mathbf{r}, \mathbf{f}_A, \mathbf{N}_A^I, \lambda)$ as the objective value of the other player that represents all human drivers, which also depends on the decisions of both players and corresponds to (12). In this case, the candidate solution derived from the best response algorithm, i.e., $(\mathbf{n}^*, \tilde{N}_{i,H}^{t*}, \hat{N}_{i,H}^{t*}, \mathbf{r}^*, \mathbf{f}_A^*, \mathbf{N}_A^{I*}, \lambda^*)$, is an ϵ-Nash equilibrium if the following conditions are satisfied:

$$\begin{cases} \pi_1(\mathbf{n}^*, \mathbf{r}^*, \mathbf{f}_A^*, \mathbf{N}_A^{I*}, \lambda^*) \geq \pi_1(\mathbf{n}^*, \mathbf{r}, \mathbf{f}_A, \mathbf{N}_A^I, \lambda) - \epsilon, & \forall \mathbf{r}, \mathbf{f}_A, \mathbf{N}_A^I, \lambda \text{ that satisfies (22)}, \\ \pi_2(\mathbf{n}^*, \mathbf{r}^*, \mathbf{f}_A^*, \mathbf{N}_A^{I*}, \lambda^*) \geq \pi_2(\mathbf{n}, \mathbf{r}^*, \mathbf{f}_A^*, \mathbf{N}_A^{I*}, \lambda^*) - \epsilon, & \forall \mathbf{n} \text{ that satisfies (13a)}. \end{cases} \tag{39}$$

where $\tilde{N}_{i,H}^{t*}$ and $\tilde{N}_{i,H}^{t*}$ can be viewed as a function of $\mathbf{n}^*$ defined through (14) and (15). At an ϵ-Nash equilibrium, each player can improve its objective value by at most ϵ by deviating from the equilibrium strategy $(\mathbf{n}^*, \tilde{N}_{i,H}^{t*}, \hat{N}_{i,H}^{t*}, \mathbf{r}^*, \mathbf{f}_A^*, \mathbf{N}_A^{I*}, \lambda^*)$, assuming that the other player adheres to the equilibrium strategy. Consequently, players are motivated to adopt the equilibrium strategy if they are indifferent to a change of ϵ. For this reason, the ϵ-Nash equilibrium defined in (39) can be viewed as an approximation to the exact Nash equilibrium, with ϵ quantifying the approximation error: the smaller ϵ is, the smaller the approximation error. Note that adopting an ϵ-Nash equilibrium implies the assumption that individual players in the game are indifferent to a small change in their objective value by up to ϵ. Thus, for each player, an ϵ-Nash equilibrium is considered to be as satisfactory as an exact Nash equilibrium. It is also worth noting that the Nash equilibrium can be considered as a special case of the ϵ-Nash equilibrium, corresponding to the case where $\epsilon = 0$.

Next, we show that there exists an ϵ so that the candidate Nash equilibrium derived from the best response algorithm, i.e., $(\mathbf{n}^*, \tilde{N}_{i,H}^{t*}, \hat{N}_{i,H}^{t*}, \mathbf{r}^*, \mathbf{f}_A^*, \mathbf{N}_A^{I*}, \lambda^*)$, is an ϵ-Nash equilibrium, and furthermore, an upper bound on ϵ can be derived by the relaxed optimization problem (28). In particular, let π_1^* denote the optimal value of (28), which can be easily derived if (28) is concave, then we have the following proposition:

Proposition 2. *Suppose that conditions for Proposition 1 holds, and assume that the best response algorithm for the two-player game (12) and (21) converges to a candidate Nash equilibrium, which is denoted as $(\mathbf{n}^*, \tilde{N}_{i,H}^{t*}, \hat{N}_{i,H}^{t*}, \mathbf{r}^*, \mathbf{f}_A^*, \mathbf{N}_A^{I*}, \lambda^*)$, then the candidate solution is an ϵ-Nash equilibrium, where ϵ satisfies:*

$$\epsilon = \pi^* - \pi_1(\mathbf{n}^*, \mathbf{r}^*, \mathbf{f}_A^*, \mathbf{N}_A^{I*}, \lambda^*). \tag{40}$$

Proof. Based on the definition (39), it suffices to show that

$$\pi_1(\mathbf{n}^*, \mathbf{r}^*, \mathbf{f}_A^*, \mathbf{N}_A^{I*}, \lambda^*) \geq \pi_1(\mathbf{n}^*, \mathbf{r}, \mathbf{f}_A, \mathbf{N}_A^I, \lambda) - \left(\pi^* - \pi_1(\mathbf{n}^*, \mathbf{r}^*, \mathbf{f}_A^*, \mathbf{N}_A^{I*})\right). \tag{41}$$

for any $\mathbf{r}, \mathbf{f}_A, \mathbf{N}_A^I$, and λ that satisfies (22), and that

$$\pi_2(\mathbf{n}^*, \mathbf{r}^*, \mathbf{f}_A^*, \mathbf{N}_A^{I*}, \lambda^*) \geq \pi_2(\mathbf{n}, \mathbf{r}^*, \mathbf{f}_A^*, \mathbf{N}_A^{I*}, \lambda^*) - \left(\pi^* - \pi_1(\mathbf{n}^*, \mathbf{r}^*, \mathbf{f}_A^*, \mathbf{N}_A^{I*})\right). \tag{42}$$

Algorithm 1 Computing approximate Nash equilibrium algorithm

Step 1: Best response to compute the candidate solution:
Input: Model parameters, initial guesses, initial states ($N^{I,t}_{i,A}, N^{I,t}_{i,H}, \hat{N}^t_{i,H}, \tilde{N}^t_{i,H}, etc.$).
Output: Candidate solution.
while *converge criteria not met* **do**

Given drivers decisions $\hat{N}^t_{i,H}, \tilde{N}^t_{i,H}$.
Solve platform profit maximization problem (21).

$$\max_{\mathbf{r},\mathbf{f}_A,\mathbf{N}^I_A,\lambda} \quad \sum_{t=1}^{T}\sum_{i=1}^{N}\sum_{j=1}^{N} r^t_{ij}\cdot\lambda^t_{ij}\cdot d_{ij} - C\cdot N_A$$

Subject to (22).

Given platform decisions $\mathbf{r}, \mathbf{f}_A, \mathbf{N}^I_A, \lambda$:
Solve driver flow (12)

$$\underset{\mathbf{n}\geq 0, N_H\geq 0}{\text{maximize}} \quad \sum_{t\in\mathcal{T}}\left(\int_0^{N^t_H} R^t_H(w)\mathrm{d}w\right) - \sum_{p\in\mathcal{P}} c_p\cdot n_p - \int_0^{N_H} F_d^{-1}(w)\mathrm{d}w$$

$$\begin{cases} \sum_{p\in\mathcal{P}} n_p = N_H, & \forall p\in P,\\ \sum_{p\in\mathcal{P}} n_p\cdot\delta^t_p = N^t_H, & \forall t\in\mathcal{T}, \forall p\in P,\\ n_p\geq 0. \end{cases}$$

end
Step 2: Compute upper bound for candidate solution and approximate ϵ:
Input: Candidate solution from **Step 1**.
Solve

$$\max_{\mathbf{r},\mathbf{f},\mathbf{N}^I,\lambda} \quad \sum_{t=1}^{T}\sum_{i=1}^{N}\sum_{j=1}^{N} r^t_{ij}\cdot\lambda^t_{ij}\cdot d_{ij} - C\cdot N_A$$

Subject to (29).

Output: Upper bound for candidate solution and the value of ϵ.

for any $\mathbf{n}$ that satisfies (13a). To prove (41), we note that it is equivalent to

$$\pi^* \geq \pi_1(\mathbf{n}^*, \mathbf{r}^*, \mathbf{f}^*_A, \mathbf{N}^{I*}_A, \lambda^*). \tag{43}$$

for any $\mathbf{n}$ that satisfies (13a). This is true because π^* is the globally optimal solution to the relaxed problem (28), which is certainly greater than $\pi_1(\mathbf{n}^*, \mathbf{r}, \mathbf{f}_A, \mathbf{N}^I_A, \lambda)$ for any $\mathbf{n}$ that satisfies (13a). To prove (42), we note that based on Proposition 1, $\mathbf{n}^*$ is the globally optimal solution to (12), thus we have:

$$\pi_2(\mathbf{n}^*, \mathbf{r}^*, \mathbf{f}^*_A, \mathbf{N}^{I*}_A, \lambda^*) \geq \pi_2(\mathbf{n}, \mathbf{r}^*, \mathbf{f}^*_A, \mathbf{N}^{I*}_A, \lambda^*) \geq \pi_2(\mathbf{n}, \mathbf{r}^*, \mathbf{f}^*_A, \mathbf{N}^{I*}_A, \lambda^*) - \left(\pi^* - \pi_1(\mathbf{n}^*, \mathbf{r}^*, \mathbf{f}^*_A, \mathbf{N}^{I*}_A)\right). \tag{44}$$

where the first inequality is due to the global optimality of $\mathbf{n}^*$, and the second inequality is because $\pi^* \geq \pi_1(\mathbf{n}^*, \mathbf{r}^*, \mathbf{f}^*_A, \mathbf{N}^{I*}_A)$. This completes the proof. □

The aforementioned procedure is summarized in Algorithm 1, where the best response algorithm is first employed to derive a candidate solution that proves to be an ϵ-Nash equilibrium, then the relaxed profit maximization problem (28) is utilized to offer an upper bound on (21) and characterize the value of ϵ.

Remark 8. The proposed algorithm presents a significant contribution in dealing with non-convex optimization problems in game-theoretic models. In particular, the profit maximization problem of the platform is non-concave, which make is difficult to obtain its globally optimal solution. In light of this our approach has been to seek a candidate solution for the original problem and then assess its proximity to the unknown globally optimal solution by establishing a theoretical upper bound on their potential gap. To this end, we have developed a relaxed version of the original problem. The relaxed problem is convex and thus allows for an efficient determination of its global optimum. Crucially, this global optimum also acts as an upper bound for the objective value of the original problem, given that the relaxed problem's optimal outcome must be at least as good as that of the original one. Hence, the objective value of the unknown globally optimal solution will be sandwiched by: (a) the objective value of the candidate solution for the original problem; and (b) the objective value of the globally optimal solution for the relaxed problem. Therefore,

by evaluating the gap between (a) and (b), we can establish an upper bound on the distance between the objective value of the candidate solution for the original problem and that of the globally optimal solution for the original problem, although the exact value of the globally optimal solution for the original problem is unknown. This bound further leads to a rigorous characterization of the approximation error (i.e., value of ϵ) for the approximate Nash equilibrium of the game.

5. Numerical simulations

This section details numerical studies conducted to validate the proposed model framework and the corresponding solution algorithm. Firstly, we will elaborate on the setup of our numerical analysis. We will then examine the performance of the proposed solution algorithm. Following this, we will explore the operational strategies of the platform with a mixed fleet and compare these with two benchmarks: one utilizing only human drivers and the other exclusively AVs, to highlight the synergistic value of combining AVs and human drivers in offering ride-hailing services.

5.1. Simulation setup

The numerical simulations are based on the recorded trip data of ride-hailing services in Manhattan on November 7, 2019, Thursday (Weekday) and November 9, 2019, Saturday (Weekend), which is procured from the New York City Taxi & Limousine Commission (TLC) (The New York City Taxi and Limousine Commission (TLC), 2019). TLC is tasked with licensing and overseeing New York City's Medallion (Yellow) taxis, for-hire vehicles, commuter vans, and paratransit services. Our focus is on data from for-hire vehicles, which encompasses trips from TNC platforms such as Uber and Lyft. The dataset provides comprehensive details for each trip, including pickup and drop-off locations, pickup time, as well as trip duration. The coordinates for these locations are given at the granularity of taxi zones, which are included within the dataset. We have consolidated all trips occurring within the Manhattan borough by the zone location IDs in the data provided. As illustrated in Fig. 2, Manhattan is segmented into six distinct zones. The color intensity of each zone corresponds to the volume of travel demand—darker shades denote areas of higher demand. We note that while our proposed method can be adapted for any number of zones, the choice of zoning strategy involves a trade-off: a higher number of zones can lead to excessively granular traffic demand between numerous origin–destination pairs, leading to many OD pair having zero (or close to zero) demand, which is unnecessary as it increases the computation time without offering any additional insights. Conversely, too few zones may result in market outcomes that lack the resolution to discern finer location details of the trips. In our study, we opt to divide the city into six zones to achieve a balance, making the analysis computationally manageable while ensuring that each origin–destination pair has substantial demand. This zoning also provides adequate geographical detail to discern demand patterns, for example, between highly urbanized core areas and less dense regions in the city.

Time is discretized into 288 intervals of $\Delta t = 5$ min each to represent a full day. It is important to note that while solving the platform's optimization problem (21), the algorithm and its corresponding upper bound are theoretically applicable for any arbitrary time horizon length T, but the computational time increases polynomially with its length. Practically, it is essential to solve (12) throughout the entire planning horizon due to strong dynamics over time. However, for (21), applying the proposed algorithm over an extensive period (e.g., 24 h) is unnecessary because the platform's control decisions predominantly affect system states in the imminent future. For example, the fleet state in the afternoon is unlikely to be influenced by the dispatch decisions made in the morning, particularly when human drivers' decisions are predetermined. Consequently, extending the operation horizon beyond a certain threshold only adds to computational time without substantially enhancing the objective value. In response to this, we implement a solution for (21) using a truncated horizon approach. Instead of determining an optimal control sequence for the entire horizon, we focus on maximizing the platform's profit within a truncated prediction horizon significantly shorter than T, such as 30–60 min. We then partially execute the derived control sequence, recede the horizon, and repeat this process until the end of the operation horizon. We will demonstrate numerically that this approach incurs only a negligible error in the computation of the optimal solution for (21).

The model parameters involved in the numerical simulation are summarized below:

$$\Theta = \{\lambda_{ij}^{0,t}, N_0, \alpha, \beta, L, w_{max}^p, \kappa, \eta, \omega_0, \omega_1, \omega_2, \xi, C, N_A\}. \tag{45}$$

The values these parameters are set as follows:

$$\begin{aligned} &N_0 = 40000, \alpha = 1.5, \beta = 1, L = 100, w_{max}^p = 15\,\text{min}, \kappa = 0.05, \\ &\eta = 0.1, \omega_0 = 2, \omega_1 = 3, \omega_2 = 4, \sigma = 4.8, \xi = 0.08, C = \$15/\text{h}, N_A = 1000. \end{aligned} \tag{46}$$

The parameter values are calibrated to ensure that the resulting solutions of the game are in close agreement with the real-world data from Manhattan City (The New York City Taxi and Limousine Commission (TLC), 2019). We employ the specific logit model, as described by Eq. (4), to capture the elastic trip demand. Here, the outside option cost c_{ij}^0 is used as an exogenous parameter that varies for each OD pair. The cost of the outside option c_{ij}^0 is assumed to be linearly dependent on the distance between nodes i and j, where longer distance indicates higher outside option cost. The scaling parameter is calibrated to so that the solution of the game is in alignment with the real-world traffic statistics data (e.g., modal share, trip volumes, etc.). The potential demand $\lambda_{ij}^{0,t}$ is adjusted to satisfy the condition $0.2 \cdot \lambda_{ij}^{0,t} \approx \lambda_{ij}^t$, where λ_{ij}^t represents the actual demand as reported in the open-source dataset. This adjustment ensures that about twenty percent of potential passengers opt for ride-hailing services (Toman et al., 2020). In this simulation study, the travel time for human drivers t_{ij}^H is estimated based on the average trip duration from the Google Map, and the

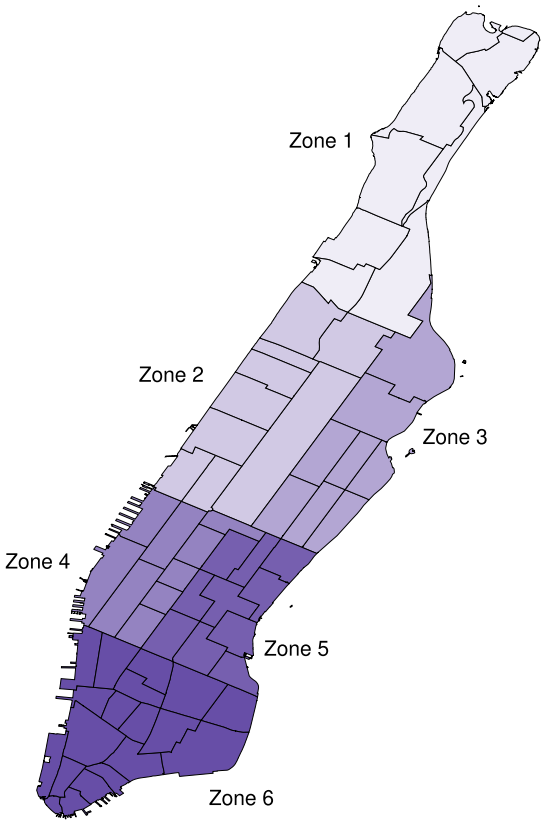


Fig. 2. Redesigned map of Manhattan City, which is segmented into six zones. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

average travel time t_{ij}^{A} for AVs between the same OD pair is assumed to be 10% longer.[7] The detailed information for trip duration is provided in Appendix A.

Remark 9. While traditional ride-hailing demand data with human-driven vehicles are accessible for several major cities (such as Chicago and New York City), ride-hailing demand data for autonomous rid-hailing service platforms, such as Waymo and Cruise, have not yet been made public. However, we clarify that the lack of AV demand data does not significantly compromise the validity of our proposed model. Our paper primarily utilizes demand data to calibrate the model, which requires to know the spatial distribution of origins and destinations within ride-hailing networks. In our mixed-fleet model, we initially calibrate the demand models using human driver data from Uber and Lyft, assuming a baseline scenario where AV operational cost is high so that it deters platform from employing AVs. This results in a special case of our model, where platform operates solely with human drivers, aligning with the real environment where human-driven vehicle data on ride-hailing demand is collected. Once the model is calibrated to this baseline scenario, it can autonomously project how demand will shift with the advent of AVs. This method, widely used in transportation planning, allows us to extrapolate from a "known case" (exclusively human drivers) to predict demand in a "new case" (with AVs), capturing the dynamics across distinct scenarios. We acknowledge that availability of additional AV data could further refine the demand model and improve its accuracy, and this is an important direction for future research once more data becomes available.

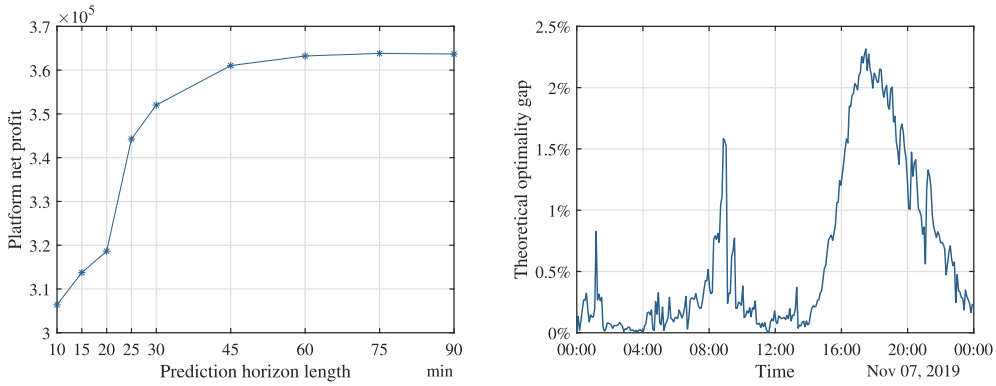
5.2. Algorithm performance

To verify the effectiveness of our solution algorithm, we conduct a numerical study using data from Manhattan. Firstly, we find that the conditions required by Proposition 1 is true not just for the precise parameter values set in the model but also for a broad

[7] This assumption is based on the observation that AVs are programmed to strictly adhere to speed limits, while human drivers frequently choose to exceed these limits, potentially achieving higher speeds. However, it is important to note that in some scenarios, AVs can achieve faster speeds than human drivers by coordinating with each other to enhance traffic flow. Despite these variations, our numerical simulation shows that either scenario yields the same qualitative results. Therefore, we present only one case in our discussion to conserve space.

Table 1
The left-hand side and right-hand side of inequality (30b).

Zones	Zone 1	Zone 2	Zone 3	Zone 4	Zone 5	Zone 6
X_i	6.2475	6.2465	6.2399	6.2443	6.2457	6.2483
$(L/w^p_{max})^2$	44.4444	44.4444	44.4444	44.4444	44.4444	44.4444



(a) Platform pure profit under varying prediction horizon lengths (b) Optimality gap of the proposed solution algorithm at distinct times of the day

Fig. 3. Platform pure profit and optimality gap of the proposed solution algorithm.

spectrum of perturbed parameter values, indicating robustness in the model's application to real-world scenarios. In particular, the values of X_i for each zone i and the right-hand side of (30b), i.e., $(L/w^p_{max})^2$, are summarized in the follow table, which clearly shows that X_i is smaller than $(L/w^p_{max})^2$ by a very large margin (see Table 1).

Secondly, as we have previously discussed, extending the prediction horizon too far into the future can result in increased computational complexity without proportionate benefits. This is because the decisions made by ride-hailing platforms tend to mainly affect the near future. Consequently, we employ a receding horizon approach, which involves solving the optimization problem at each time step for a truncated horizon. To verify the effectiveness of this strategy, we calculate the optimal profit of the platform over a range of distinct prediction horizon lengths, from 10 to 90 min. As depicted in Fig. 3(a), we observe that extending the prediction horizon from 30 to 45 min improves the profit of the platform by 2.4%, while further extending the prediction horizon from 45 to 90 min leads to an incremental increase in cumulative platform profit by just 0.727%. This suggests that the benefits of forecasting beyond 45 min are minimal, indicating that a 45-minute horizon is adequate for the platform to make informed decisions about pricing and vehicle positioning. Therefore, we choose our prediction horizon to be 45 min in the simulation section of the paper.

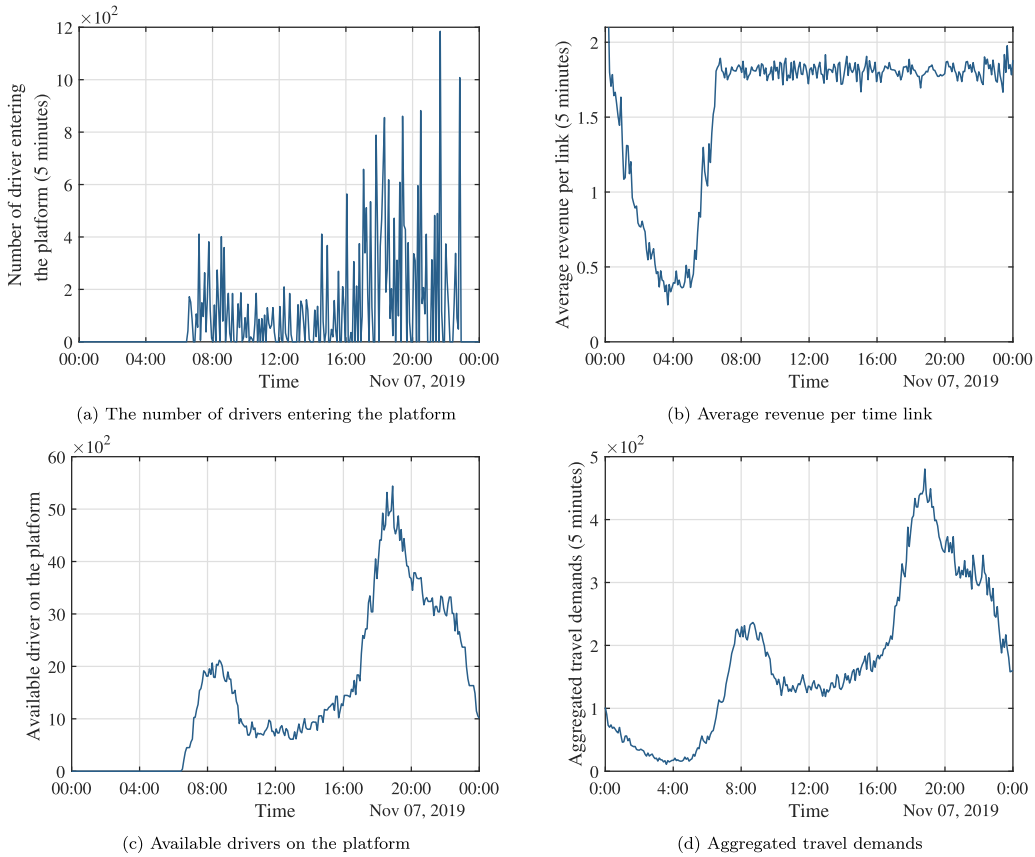
Thirdly, we employ the interior-point algorithm to iteratively solve the best response problems (21) and (12) until convergence. Subsequently, drawing upon Algorithm 1, we address the relaxed problem (28) to establish an upper bound for the optimality gap between the optimal value of the relaxed problem (28) and the unknown globally optimal value of the original optimization problem (21). We assess the optimality of the obtained solutions using this theoretical upper bound as a benchmark. Fig. 3(b) depicts the percentage gap between the optimal solutions of the original problem and its relaxed counterpart. Our findings reveal that the solutions produced by the best-response algorithm are highly accurate, exhibiting a consistently negligible optimality gap that ranges from 0% to 2.2% across different time intervals, with the bound capped at 2.2% when expressed as a percentage. Note that since the optimization problem for the human drivers is convex, it indicates that the value of ϵ is also upper bounded at 2.2% when expressed as a percentage value. This indicates that the solutions we have identified are almost globally optimal for the TNC platform, and the resulting decisions of the platform and drivers altogether constitutes a high-quality approximate Nash equilibrium of the game. Note that the significance of such a small optimality gap is that it implies the platform has the assurance of near-optimality when deploying our solutions, even in the face of non-convexity, where such assurances are typically challenging to obtain.

5.3. Analysis of the market equilibrium

5.3.1. The case of weekday demand data

This subsection presents the simulation results corresponding to the case of the mixed fleet and analyze the implications of the corresponding market equilibrium using the demand data on weekday. We follow Algorithm 1 outlined in Section 4 to derive the ϵ Nash Equilibrium of the game. The simulation results are shown in Figs. 4(a)–5(f), and several interesting observations are discussed below.

Figs. 4(a) to 4(c) presents the optimal drivers working choice and schedule obtained from the model (12). Fig. 4(a) illustrates that in the case of mixed fleet, the majority of human drivers that enter the platform operate during the day, whereas no human



(a) The number of drivers entering the platform

(b) Average revenue per time link

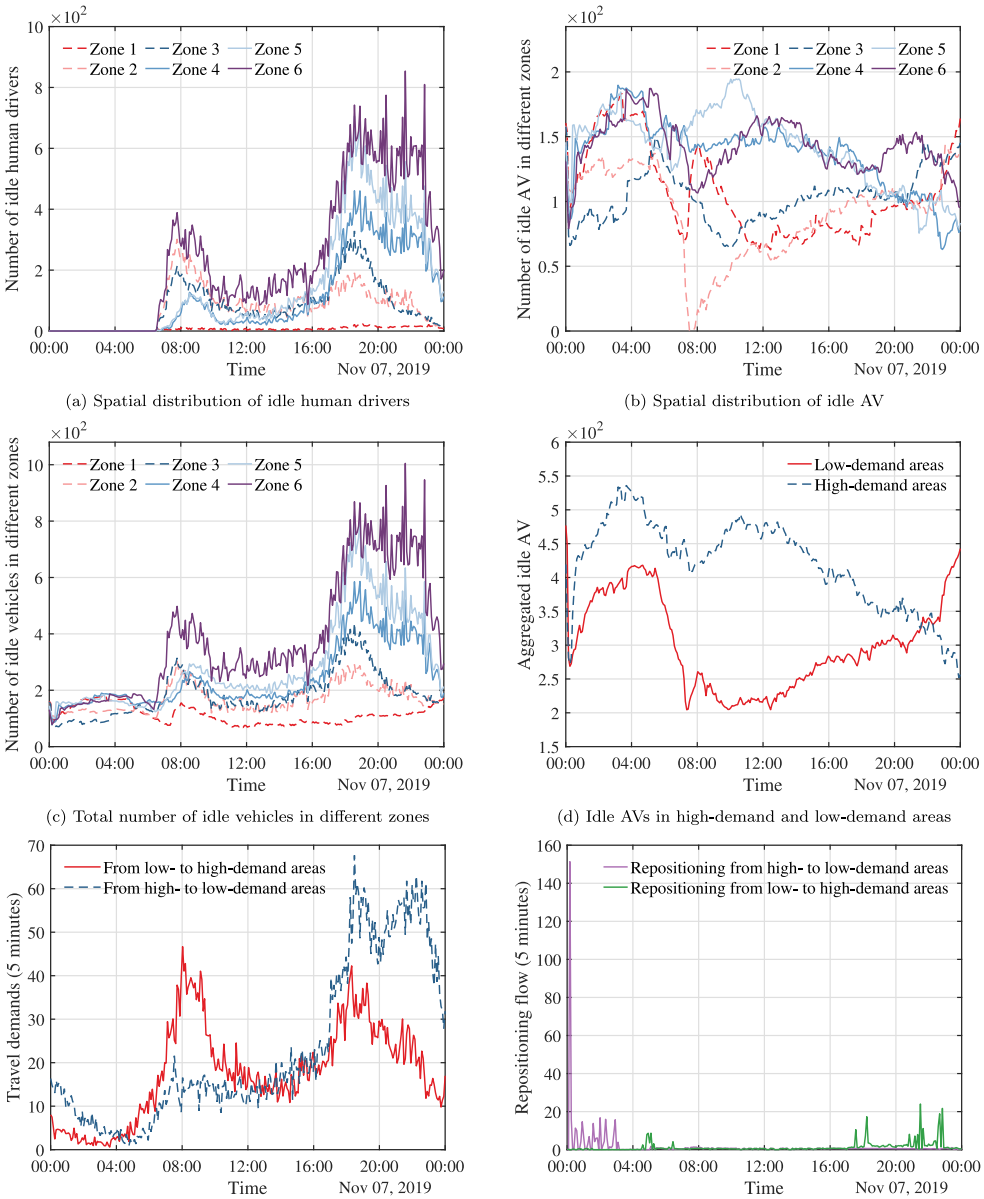
(c) Available drivers on the platform

(d) Aggregated travel demands

Fig. 4. Optimal drivers working choice and schedule and aggregated travel demands.

drivers is willing to work during 00:00-06:00 due to the low passenger demand. This gap in human-driven services is filled by AVs, which are operational 24/7, thereby ensuring that demand is met throughout the night. Fig. 4(b) presents the average revenue of human drivers for each 5-minute intervals at the market equilibrium. The revenue during night time is lower than that during the day, whereas the revenue during the day is relative steady. This pattern is expected as the potential for higher earnings during specific daytime intervals draws more drivers, creating competition and consequently reducing the revenue. Fig. 4(c) illustrates the variability in the number of human drivers on the platform throughout a full day. A pronounced peak in driver numbers is observed during the morning and afternoon rush hours, contrasting with the complete absence of drivers at night. This fluctuation closely aligns with the demand patterns depicted in Fig. 4(d), indicating that human drivers' work schedules are in sync with demand fluctuations, despite a saturation of drivers during peak times. Moreover, in the quieter hours after midnight (1:00–5:00), when demand plummets, the incentive for human drivers to work diminishes due to lower potential earnings. This behavior reinforces the notion that human drivers act as a supplemental labor force for the ride-hailing platform. During low-demand periods, such as late at night, human drivers tend to withdraw from the platform, leaving AVs to cater to the residual customers. Conversely, during periods of high demand, human drivers are more inclined to provide services, thereby supporting the platform in meeting peak demand.

Figs. 5(a) and 5(b) showcase the spatio-temporal distribution of idle human drivers and AV across various city zones at different times of the day. Fig. 5(c) aggregates these figures to illustrate the total count of idle vehicles, combining both idle human drivers and idle AVs. From these figures, a clear correlation emerges between the distribution of idle human drivers and passenger demand patterns, whereas the distribution of idle AVs significantly deviates from this trend. Specifically, Fig. 5(a) reveals that human drivers are strategically positioned in areas with high demand, adjusting spatially to match passenger requests. Temporally, the quantity of idle human drivers fluctuates in tandem with demand, peaking during the morning and afternoon rush hours. Conversely, during nighttime hours, there is a marked absence of human drivers. In contrast, Fig. 5(b) indicates a less predictable pattern for idle AVs. There is no direct correlation between high-demand areas and the presence of idle AVs. Additionally, AVs maintain a considerable presence throughout the night, demonstrating their non-reliance on immediate demand signals for positioning. These observations suggest that human drivers are responsive to fast-changing demand signals, aligning their availability with passenger needs from one minute to another. On the other hand, the distribution of AVs seems to be governed by a different set of operational strategies, potentially focusing on broader coverage and readiness to serve future demand rather than immediate passenger requests. This disparity highlights the distinct operational dynamics between human drivers and AVs within the ride-hailing ecosystem.



(a) Spatial distribution of idle human drivers

(b) Spatial distribution of idle AV

(c) Total number of idle vehicles in different zones

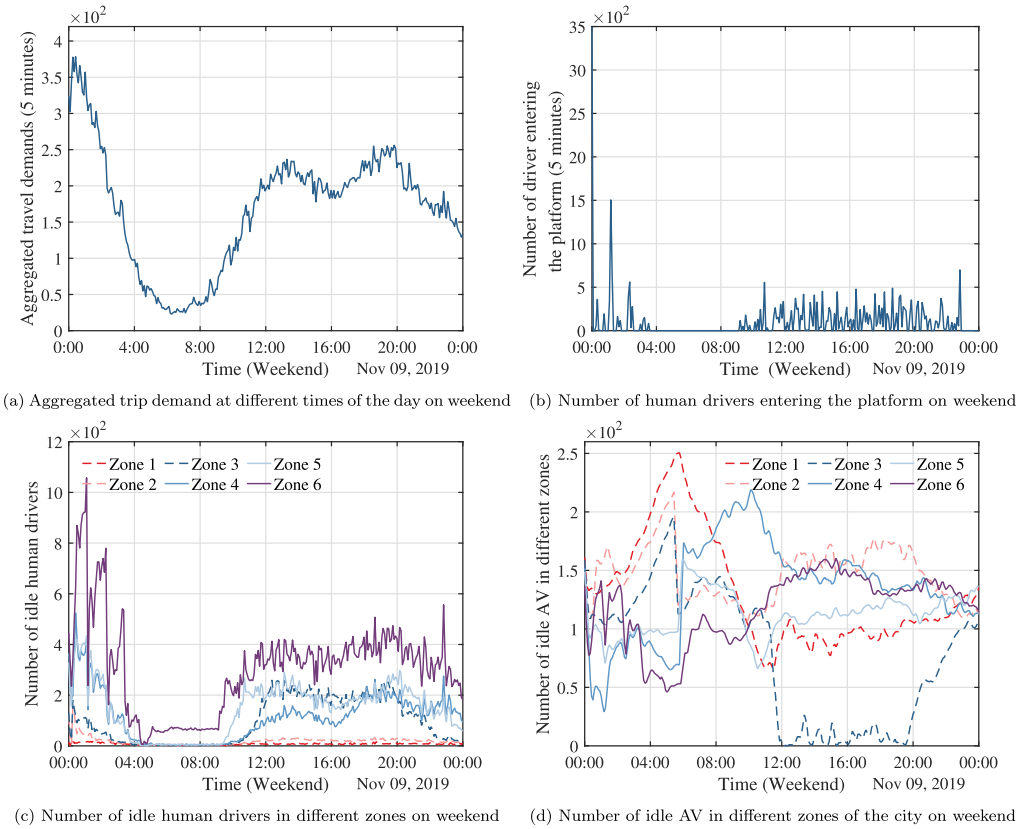
(d) Idle AVs in high-demand and low-demand areas

(e) Aggregated demand between high-demand and low-demand areas

(f) Reposition flow of AVs between high-demand and low-demand areas

Fig. 5. Market outcomes under weekday demand data.

To elucidate the distinct operational dynamics and to unravel the spatio-temporal patterns of idle AVs, we have further segmented Manhattan into two areas based on travel demand. Specifically, data shows that approximately 69.72% of trips originate from Zones 4, 5, and 6. Consequently, we define a low-demand area comprising Zones 1, 2, and 3, and a high-demand area encompassing Zones 4, 5, and 6. Fig. 5(d) illustrates the spatio-temporal distribution of idle AVs within these two areas. After consolidation, a pronounced pattern in the dynamics of idle AV distribution becomes evident. Specifically, during daytime hours (07:00–24:00), the high-demand area shows a decreasing trend in idle AV numbers, reaching the lowest number at midnight. Conversely, the low-demand area exhibits an increasing trend, peaking at midnight. This trend closely correlates with the OD distribution of passenger demand, as depicted in Fig. 5(e). Specifically, we observe a significant flow of passengers from residential (low-demand) areas to commercial (high-demand) areas in the morning, and a reverse flow late at night. This passenger demand pattern strongly influences AV distribution. For example, with fewer human drivers available late at night, AVs accommodate the substantial number of passengers traveling from high-demand to low-demand areas, leading to an increase in idle AVs in residential zones. Additionally, to efficiently serve the morning rush, the platform proactively redistributes idle AVs within low-demand areas overnight, ensuring



(a) Aggregated trip demand at different times of the day on weekend (b) Number of human drivers entering the platform on weekend

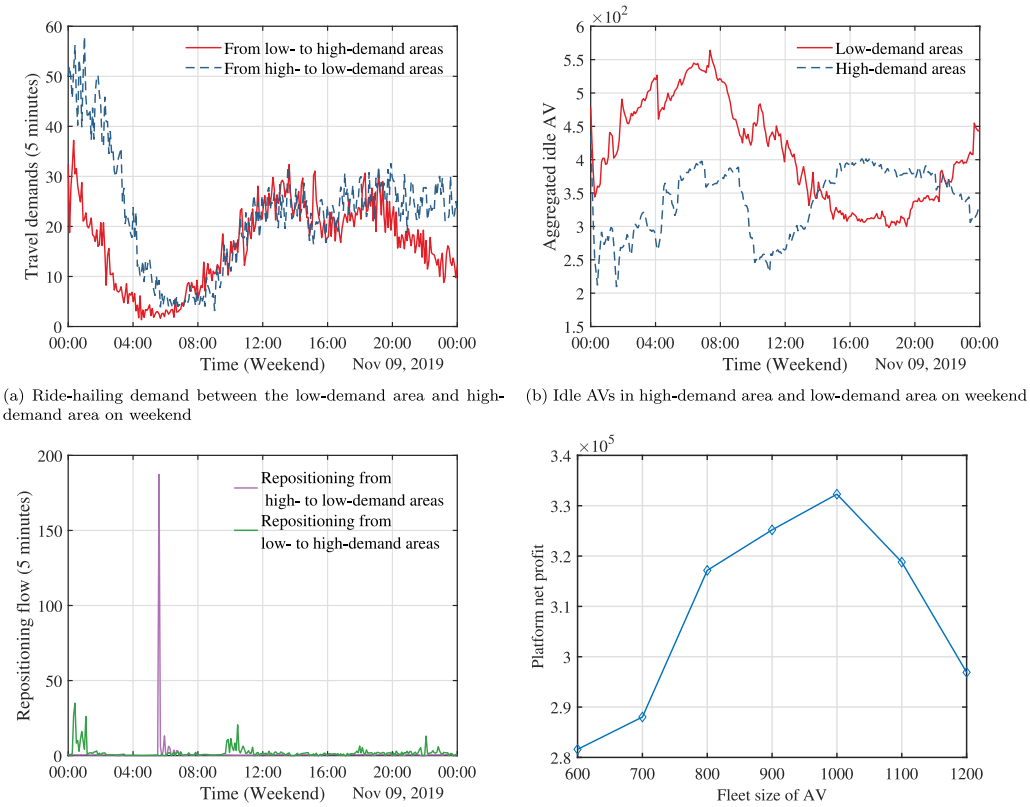
(c) Number of idle human drivers in different zones on weekend (d) Number of idle AV in different zones of the city on weekend

Fig. 6. Market outcomes under weekend demand data - Part I.

availability for the morning commute. This strategy is further depicted in Fig. 5(f), which shows the repositioning flow of AVs. This analysis reveals a complementary aspect of human drivers and AVs: human drivers are closely aligned with the immediate, minute-to-minute fluctuations in passenger demand, enabling them to effectively meet the surge during peak demand windows, whereas AVs are more attuned to the extended, daily cycles of passenger movements throughout different city zones, from daytime to nighttime. This synergy ensures that the ride-hailing platform can effectively serve demand fluctuations of different temporal resolutions throughout the day.

5.3.2. The case of weekend demand data

Note that the insights discussed thus far are based on data from a typical weekday. To assess the robustness of these insights in the face of varying demand patterns, we conducted additional simulations using weekend travel demand data. The results of these weekend simulations are presented in Figs. 6(a) through 7(c). Interestingly, we notice a number of differences when comparing weekend results to those of weekdays. In particular, during weekends, there is a spike in ride-hailing demand at night (e.g., between 0:00 and 2:00 in Fig. 6(a)), which could be intuitively linked to people engaging in late-night entertainment activities. This nocturnal traffic draws a significant number of human drivers to opt for night shifts (Fig. 6(b) and Fig. 6(c)). This pattern diverges from the weekday scenario, where midnight demand is predominantly served by AVs, while human drivers primarily operate during daylight hours (Fig. 5(a)). Moreover, when we categorize the zones into high-demand and low-demand areas, similar to our previous analysis for weekdays, we observe a substantial flow of travel from high to low-demand areas at midnight during weekends (Fig. 7(a)). This trend likely reflects the movement of people from central entertainment zones (i.e., high-demand areas) back to residential areas (i.e., low-demand areas) late at night, paralleling the weekday pattern. However, a key difference is observed in the morning: unlike weekdays, there is no significant influx of traffic from low-demand (residential) to high-demand (city core) areas, which can be attributed to the absence of weekday commuting patterns. Interestingly, these patterns also influence the spatial and temporal distribution of idle AVs. On weekends, we still see a concentration of idle AVs in low-demand areas during the late-night hours (Fig. 7(b)), as a significant number of passengers are transported to these regions and then become idle. In contrast, due to the absence of a morning rush from residential to commercial areas, we do not observe a significant AV repositioning flow from high-demand to low-demand areas at midnight (Fig. 7(c)), nor is there a substantial number of idle AVs in high-demand zones in the morning (Fig. 7(b)). This is a departure from the weekday scenario and is directly related to the lack of morning commuting traffic on weekends. Overall, our findings still underscore the complementary roles of human drivers and AVs: human drivers adapt swiftly



(a) Ride-hailing demand between the low-demand area and high-demand area on weekend

(b) Idle AVs in high-demand area and low-demand area on weekend

(c) Reposition flow of AVs between high-demand and low-demand areas on weekend

(d) The profit of the platform under different AV fleet size using the weekday demand data

Fig. 7. Market outcomes under weekend demand data - Part II and optimal AV fleet size.

to immediate, short-term fluctuations in passenger demand, while AVs appear to be more aligned with the broader, daily passenger movement across different city zones. This synergy between human drivers and AVs is once again evident during weekends, even under a different demand pattern.

Remark 10. In our analysis, we maintained a constant number of AVs, denoted as $N_A = 1000$. However, in practice, the TNC platform may adjust the size of the AV fleet over the long term to optimize profits. To examine the effect of AV fleet size on platform profit, we replicated the simulation for various fleet sizes. Fig. 7(d) illustrates the relationship between platform profit and N_A. We observe that profit peaks at a certain intermediate fleet size, while it diminishes when the fleet size is either too large or too small. In real-world scenarios, TNC platforms could determine the optimal AV fleet size using a grid search, informed by the mathematical model proposed in this paper.

5.4. Comparing with benchmark cases

5.4.1. The case of weekday demand data

To further elucidate the benefits of a mixed fleet within a ride-hailing platform, we contrast the mixed fleet model against two benchmarks using the demand data on weekday, including: (a) A platform operating exclusively with for-hire human drivers; (b) A platform operating exclusively with AVs. For the first benchmark (hereafter referred to as Benchmark A) - a human driver-only model - we utilize a game-theoretic framework. Here, one player represents the profit-maximizing platform, which solves the optimization problem in Eq. (21). The second player is a virtual representation of the collective decisions made by self-interested individual human drivers, which is modeled by solving Eq. (12). To tailor the scenario to a human driver-only case, we set the AV fleet size to zero, $N_A = 0$. We also assume the absence of a maximum allowed waiting time, especially at night when driver supply is generally insufficient, making it challenging or infeasible to maintain guaranteed service levels. For the second benchmark (hereafter referred to as Benchmark B) - an AV-only scenario - the problem simplifies to a single-agent optimization problem for the platform, represented by Eq. (21). Here, the number of human drivers is set to zero. To ensure a fair comparison, we maintain the same potential trip demands $\lambda_{ij}^{0,t}$ and operational costs C for AVs. The AV fleet size in this case is determined through an exhaustive search across all possible fleet sizes to find the optimal one that maximizes the platform's profit.

Fig. 8(a) delineates the total number of vehicles in each of the three scenarios - mixed fleet, human drivers only (Benchmark A), and AVs only (Benchmark B). In the mixed fleet scenario, the total fleet size is invariably larger than that of Benchmark A. This increase aligns with expectations since the integration of AVs, which typically have lower operational costs, allows the platform to expand the total fleet size to enhance service offerings. Conversely, the number of AVs within the mixed fleet is less than in the AV-only scenario (Benchmark B). This is rational since the platform in a mixed fleet has the flexibility of employing both AVs and for-hire human drivers, diminishing the necessity for keeping a larger number of AVs in the fleet. Idle vehicle dynamics are depicted in Figs. 8(b) and 8(c) for human drivers only and AVs only, respectively. For Benchmark A, idle human drivers are scarce at night and align closely with the spatio-temporal demand pattern during the day. This correlation results in extended waiting times, particularly during off-peak periods such as afternoons and nights, as shown in Fig. 8(d).[8] The AV-only case (Benchmark B) presents a distinct pattern where there is an abundance of idle vehicles during off-peak times, but a deficiency during peak hours, leading to prolonged wait times during those periods, as seen in Fig. 8(e). When comparing the mixed fleet with the two benchmarks, it becomes evident that the average waiting time is substantially smaller (see Fig. 8(f)) and the average trip cost is generally lower (see Fig. 9(a)[9]) in the mixed fleet scenario. Furthermore, with respect to the platform's profitability, the deployment of a mixed fleet contributes to improved profit for the platform, particularly in contrast with Benchmark B, as illustrated in Fig. 9(b). Overall, the above analysis indicate that the synergistic integration of AVs and human drivers plays a significant role in addressing demand variability, optimizing the supply–demand balance across different times and locations. The mixed fleet model, therefore, demonstrates a clear advantage in reducing passenger waiting times, reducing overall travel cost, and increasing the platform profit.

Remark 11. Comparing the strengths and weaknesses of purely AV fleets and purely human-driven fleets is intriguing, as it sheds light on their potential synergy and why a mixed fleet may offer superior performance. Notably, human drivers can adjust to real-time demand fluctuations. Thus, the advantage of a human-driven fleet is in its ability to scale supply during peak hours (e.g., 8 am-10 am and 6 pm−8 pm) in response to increased demand, which facilitates a timely service and maximizes profits for the ride-hailing platform (Figs. 8(f) and 9(b)). However, the drawback is a significant decline in service quality during off-peak hours (e.g., 00:00–04:00), leading to longer waiting times and reduced profits. Conversely, the strengths and weaknesses of AVs are exactly opposed. The platform employs a fixed number of AVs, resulting in an oversupply during off-peak hours (e.g., 00:00–04:00), which ensures prompt service and boosts profits, but an under-supply of AVs during the peak hours (e.g., 8 am-10 am and 6 pm−8 pm), causing longer passenger wait times and reduced platform profit (Figs. 8(f) and 9(b)). This contrast explains why combining these two types of fleets can significantly enhance performance. By capitalizing on the strengths of both models, a mixed fleet can decrease wait times and enhance profits throughout the entire day, addressing both the peak and off-peak periods.

Aside from the advantages of reduced passenger wait times and increased platform profits, we also wish to highlight the broader impacts of AV integration into the ride-hailing market, particularly regarding urban planning, sustainability, and social equity. Firstly, our research suggests that deploying AVs will result in distinct spatio-temporal vehicle movement patterns, e.g., the relocation of AVs from high-demand areas to low-demand areas late at night (Fig. 5(f)). This shift implies a need for urban planning adaptations, such as reconsidering parking space allocation to accommodate the parking requirements of autonomous ride-hailing vehicles in low-demand areas at night. Therefore, investigating the optimal parking provision for a mixed ride-hailing fleet presents an intriguing future research avenue. Secondly, our findings indicate that the incorporation of AVs could expand the overall fleet size of TNC platforms (Fig. 8(a)), potentially leading to increased emissions and traffic congestion. This highlights the necessity for further studies on vehicle electrification or the implementation of congestion-mitigating regulatory policies to manage these potential negative impacts. Thirdly, the spatial distribution of human drivers tends to be disproportionately concentrated in high-demand areas (Fig. 5(a)), which may exacerbate transport inequity, whereas the introduction of AVs could help lessen this disparity, particularly during off-peak times (e.g., 00:00–04:00) when the number of idle vehicles in low-demand areas may surpass those in high-demand regions (Fig. 5(c)). Identifying and addressing transport inequity through the lens of AV deployment is yet another critical direction for future investigation. Overall, our research provides an initial exploration into these broader implications, aiming to spur further studies in the field, while a detailed analysis remains beyond the current scope of our work.

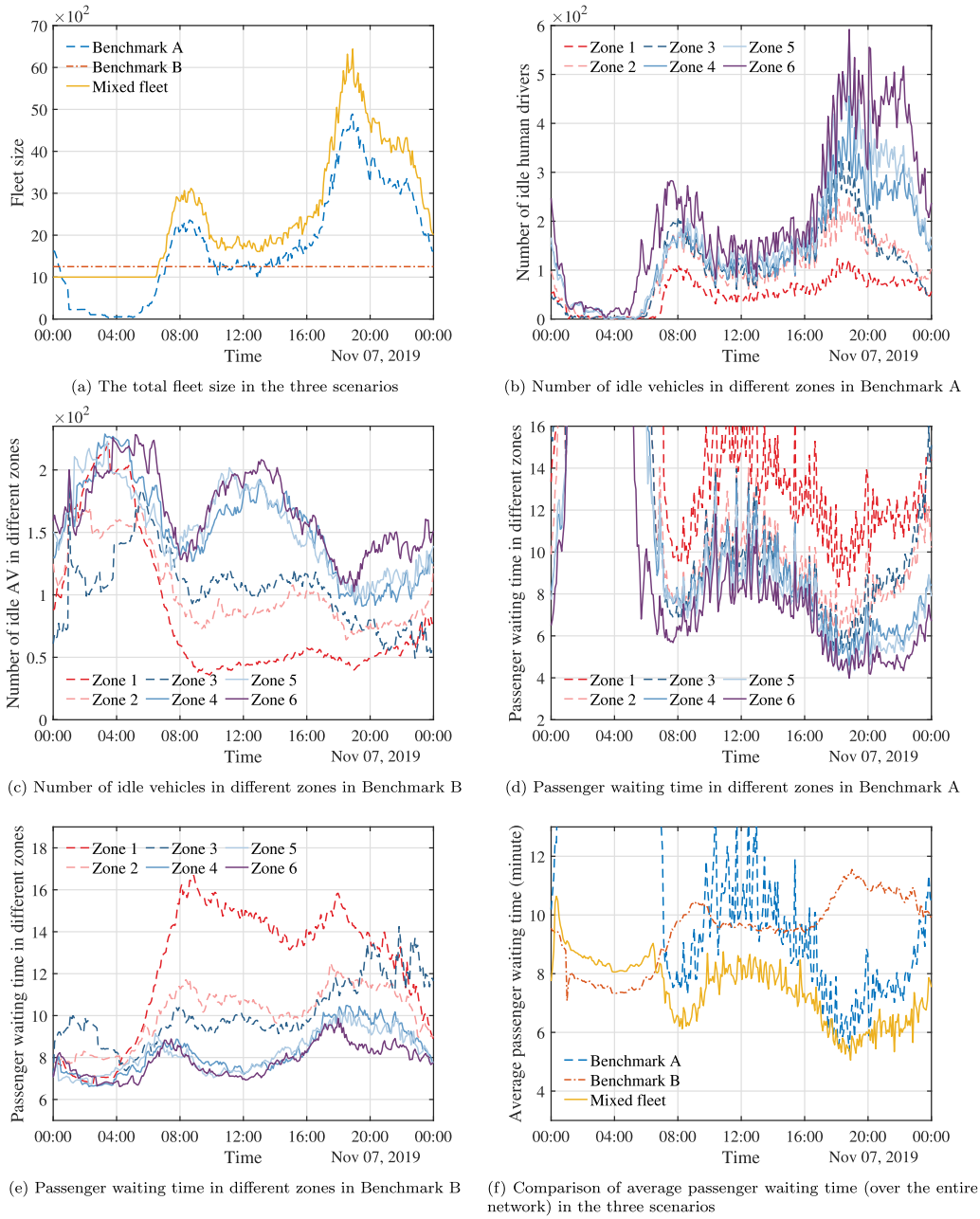
5.4.2. The case of weekend demand data

To validate the robustness of the above insights under distinct demand patterns, we repeat the simulation in Section 5.4.1 using the weekend demand data. In particular, we use the recorded trip data of ride-hailing services in Manhattan on November 9, 2019, Saturday, and contrast the mixed fleet model against two benchmarks, including: (a) A platform operating exclusively with for-hire human drivers (Benchmark A); (b) A platform operating exclusively with AVs (Benchmark B). The simulation results are shown in Figs. 10(a) through 11(f).

Fig. 10(a) delineates the total number of vehicles in each of the three scenarios. Similar to the weekday case, in the mixed fleet scenario, the total fleet size is invariably larger than that of Benchmark A, while the number of AVs within the mixed fleet is less than in the AV-only scenario. Idle vehicle dynamics are depicted in Figs. 10(b) and 11(a) for human drivers only and AVs only, respectively. For Benchmark A, we see a significant presence of idle human drivers at night, which is distinct from the results in the

[8] The waiting time at night sometimes exceeds 100 min. Therefore, to allow the daytime pattern to be discernible, we truncate the scale of the vertical axis in Fig. 8(d) at 16 (min).

[9] The average trip cost at night occasionally surpasses $80 per trip. Consequently, to ensure that the daytime cost pattern remains visible, we have truncated the scale of the vertical axis at $40 per trip in Fig. 9(a).



(a) The total fleet size in the three scenarios

(b) Number of idle vehicles in different zones in Benchmark A

(c) Number of idle vehicles in different zones in Benchmark B

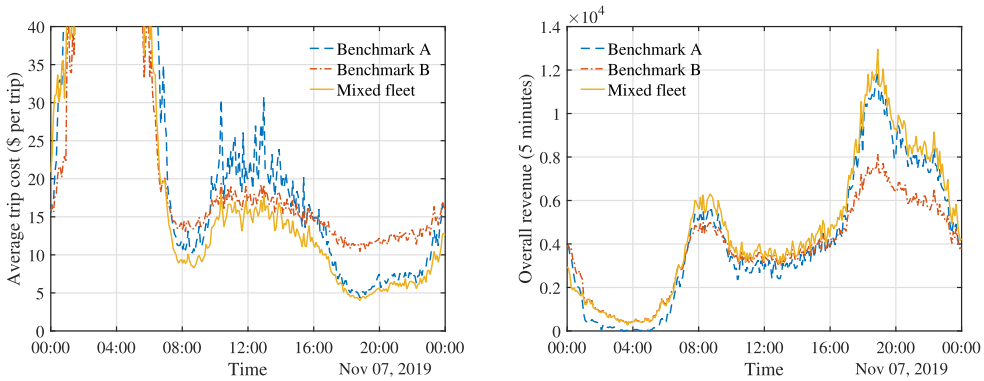
(d) Passenger waiting time in different zones in Benchmark A

(e) Passenger waiting time in different zones in Benchmark B

(f) Comparison of average passenger waiting time (over the entire network) in the three scenarios

Fig. 8. Comparing with benchmark cases under weekday demand data - Part I.

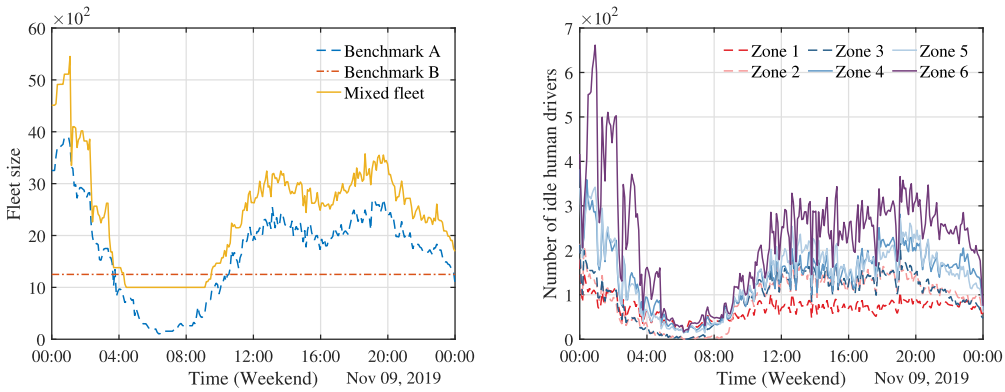
weekday case, but it is consistent with the intuition that the temporal and spatial distribution of idle human drivers aligns closely with the spatio-temporal demand pattern throughout the day. Similar to the weekday case, this correlation also results in extended waiting times during off-peak periods, as shown in Fig. 11(b). Conversely, the AV-only case (Benchmark B) presents a distinct pattern where there is a deficiency during peak hours, leading to prolonged wait times during those periods, as seen in Fig. 11(c). When comparing the mixed fleet with the two benchmarks, it becomes evident that the average waiting time is substantially smaller (see Fig. 11(d)), the average trip cost is generally lower (see Fig. 11(e)), and the platform profit is higher (see Fig. 11(f)) in the mixed fleet scenario. Overall, the above analysis indicates that the synergistic integration of AVs and human drivers plays a significant role in addressing demand variability, optimizing the supply–demand balance across different times and locations, thus it can benefit the overall system by reducing passenger waiting times, reducing overall travel cost, and increasing the platform profit. This set of simulation results confirms that the insights and conclusions obtained in the weekday case also hold in the context of weekend demand data.



(a) Comparison of average trip cost (over the entire network) in the three scenarios

(b) Comparison of platform profit (over the entire network) in the three scenarios

Fig. 9. Comparing with benchmark cases under weekday demand data - Part II.



(a) The total fleet size on weekend in the three scenarios

(b) Number of idle vehicles on weekend in different zones in Benchmark A

Fig. 10. Comparing with benchmark cases under weekend demand data - Part I.

6. Extended model: the case of traffic congestion

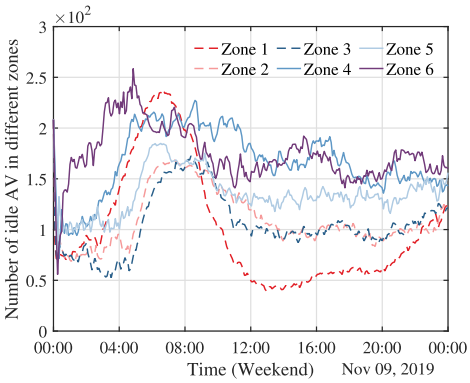
The proposed mathematical model is predicated on a simplified setting where the impact of vehicle flows from ride-hailing services on traffic congestion does not significantly affect vehicle dynamics. This assumption holds when the mixed fleet constitutes only a fraction of the overall traffic volume, and the time granularity of our model is relatively coarse (e.g., 5 min) compared to the transient effects of ride-hailing vehicles on traffic congestion (typically less than 5 min per trip). In this section, we refine the proposed model to account for the impacts of ride-hailing fleets on traffic congestion.

Note that when the impacts of ride-hailing fleet on traffic congestion is considered in a dynamic model, it is much more complex to characterize the arrival dynamics of the vehicles since it not only depends on departure time but also congestion levels along the way. To address this challenge, we represent ride-hailing trips using another time-expanded network. In particular, we define (i, j, τ) as an extended OD pair, which corresponds to a trip from zone i to zone j at departure time τ. A time-location pair (τ, k) is defined as a link, indicating the vehicle is in zone k at the beginning of time step τ. For each trip (i, j, τ), its path consists of a sequence of such links. Suppose a path m for trip (i, j, τ) sequentially passes through zone i, k, and then j. This path can be denoted as

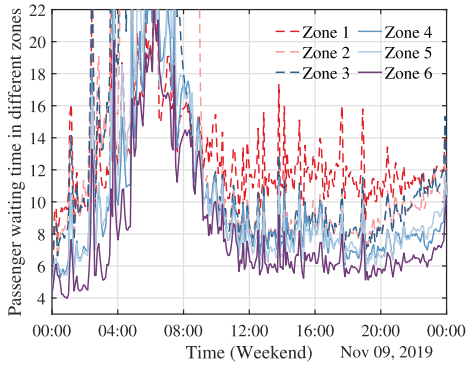
$$p^m_{ij\tau} = \left\{(\tau, i), \dots, (\tau^{mi}_{ij\tau}, i), (\tau^{mi}_{ij\tau} + 1, k), \dots, (\tau^{mk}_{ij\tau}, k), (\tau^{mk}_{ij\tau} + 1, j), \dots, (\tau^{mj}_{ij\tau}, j)\right\}, \tag{47}$$

where $\tau^{mk}_{ij\tau}$ is the last time step at the beginning of which a vehicle on path $p^m_{ij\tau}$ is staying in zone k (i.e., after this time step it enters another zone). Therefore, any vehicle going along path $p^m_{ij\tau}$ will exit from zone i and enter zone k in between $[\tau^{mi}_{ij\tau}, \tau^{mi}_{ij\tau} + 1)$, exit from zone k and enter zone j in between $[\tau^{mk}_{ij\tau}, \tau^{mk}_{ij\tau} + 1)$, and finally reach its destination in zone j in between $[\tau^{mj}_{ij\tau}, \tau^{mj}_{ij\tau} + 1)$.

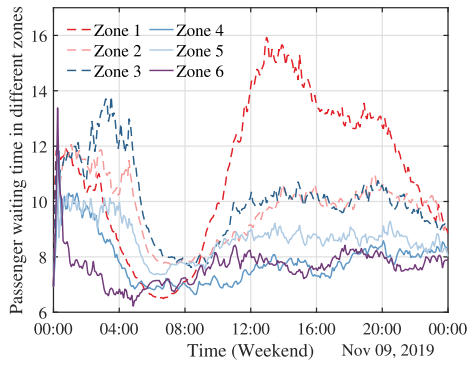
To characterize the vehicle arrival dynamics, we introduce a binary variable $\phi^m_{ij\tau}$, which is 1 if path $p^m_{ij\tau}$ is chosen for an (i, j, τ) trip and 0 otherwise. Given that all vehicles for a specific time-expanded OD pair shall arrive at the same time, it is natural to



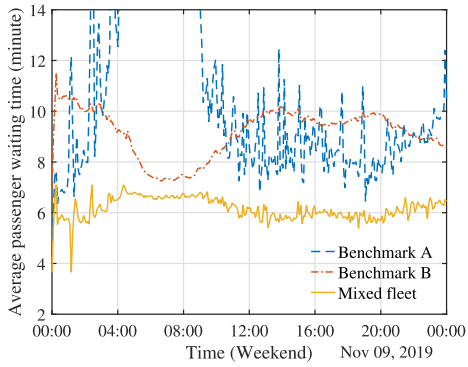
(a) Number of idle vehicles in different zones on weekend in Benchmark B



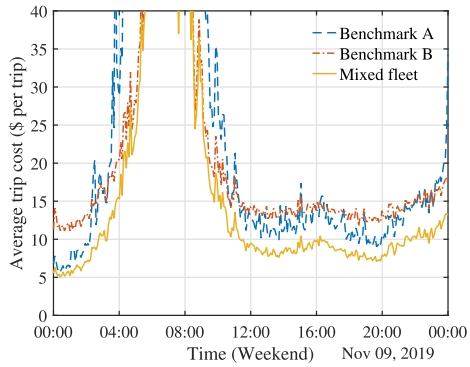
(b) Passenger waiting time on weekend in different zones in Benchmark A



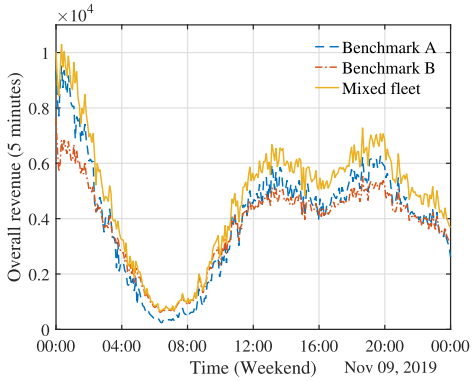
(c) Passenger waiting time in different zones on weekend in Benchmark B



(d) Comparison of average passenger waiting time on weekend (over the entire network) in the three scenarios



(e) Comparison of average trip cost on weekend (over the entire network) in the three scenarios



(f) Comparison of platform profit on weekend (over the entire network) in the three scenarios

Fig. 11. Comparing with benchmark cases under weekend demand data - Part II.

assume that only one path will be used, i.e.,

$$\sum_m \phi^m_{ij\tau} = 1. \tag{48}$$

Furthermore, we denote $g_{ij\tau} = g^A_{ij\tau} + g^H_{ij\tau}$ as the total vehicle flow that intends to travel from zone i to zone j and start the trip at time τ, including both AVs $g^A_{ij\tau}$ and human driven vehicles $g^H_{ij\tau}$. We further denote $g^m_{ij\tau}$ as the flow along path $p^m_{ij\tau}$. The path flow $g^m_{ij\tau}$ will relate to $g_{ij\tau}$ through the following conservation conditions:

$$g^m_{ij\tau} = \phi^m_{ij\tau} g_{ij\tau}. \tag{49}$$

Let $\theta^{\tau,k}_{ij\tau,m}$ be an indicator that is 1 if link (τ,k) is on path $p^m_{ij\tau}$ and 0 otherwise. The total flow in zone k at time step τ is thus obtained by

$$A_{\tau,k} = \sum_{ij\tau}\sum_{m}\theta^{\tau,k}_{ij\tau,m}\phi^m_{ij\tau}g_{ij\tau}. \tag{50}$$

This cumulative flow affects the real-time speed in each zone via a decreasing function $v_{\tau,k} = F_v(A_{\tau,k})$, and consequently the travel distance during each time step. As previously mentioned, if a vehicle opts for path $p^m_{ij\tau}$, it will exit from zone i and enter zone k within the interval $[\tau^{mi}_{ij\tau}, \tau^{mi}_{ij\tau}+1)$. To ensure the actual exit time falls into this interval, we use trips from zone i to zone j as an example, assuming that a trip from i to zone j will traverse zone i, zone k and zone j sequentially (note that this assumption does not impose any loss of generality because if there are more intermediate zones, we can add more logical constraints correspondingly). In this case, we have the following logical constraints:

$$\sum_{t=\tau}^{\tau^{mi}_{ij\tau}}\sum_{s=1}^{N}\theta^{\tau,s}_{ij\tau,m}v_{\tau,s}\Delta t - L^{mi}_{ij\tau} \le M(1-\phi^m_{ij\tau}), \tag{51a}$$

$$\sum_{\tau=t}^{\tau^{mi}_{ij\tau}+1}\sum_{s=1}^{N}\theta^{\tau,s}_{ij\tau,m}v_{\tau,s}\Delta t - L^{mi}_{ij\tau} \ge M(\phi^m_{ij\tau}-1), \tag{51b}$$

$$\sum_{\tau=t}^{\tau^{mk}_{ij\tau}}\sum_{s=1}^{N}\theta^{\tau,s}_{ij\tau,m}v_{\tau,s}\Delta t - (L^{mi}_{ij\tau}+L^{mk}_{ij\tau}) \le M(1-\phi^m_{ij\tau}), \tag{51c}$$

$$\sum_{\tau=t}^{\tau^{mk}_{ij\tau}+1}\sum_{s=1}^{N}\theta^{\tau,s}_{ij\tau,m}v_{\tau,s}\Delta t - (L^{mi}_{ij\tau}+L^{mk}_{ij\tau}) \ge M(\phi^m_{ij\tau}-1), \tag{51d}$$

$$\sum_{\tau=t}^{\tau^{mj}_{ij\tau}}\sum_{s=1}^{N}\theta^{\tau,s}_{ij\tau,m}v_{\tau,s}\Delta t - (L^{mi}_{ij\tau}+L^{mk}_{ij\tau}+L^{mj}_{ij\tau}) \le M(1-\phi^m_{ij\tau}), \tag{51e}$$

$$\sum_{\tau=t}^{\tau^{mj}_{ij\tau}+1}\sum_{s=1}^{N}\theta^{\tau,s}_{ij\tau,m}v_{\tau,s}\Delta t - (L^{mi}_{ij\tau}+L^{mk}_{ij\tau}+L^{mj}_{ij\tau}) \ge M(\phi^m_{ij\tau}-1), \tag{51f}$$

where M is a sufficiently large constant, $L^{mk}_{ij\tau}$ is the average travel distance in zone k along path $p^m_{ij\tau}$, and thus the total travel distance of path $p^m_{ij\tau}$ is $L^{mi}_{ij\tau}+L^{mk}_{ij\tau}+L^{mj}_{ij\tau}$. In the above logical constraints, (51a) ensures that the vehicles still stays in zone i at time $\tau^{mi}_{ij\tau}$ (because the travel distance is smaller than $L^{mi}_{ij\tau}$), while (51b) indicates that it leaves zone i at time $\tau^{mi}_{ij\tau}+1$ (because the travel distance is larger than $L^{mi}_{ij\tau}$). (51c) ensures that the vehicles still stays in zone k at time $\tau^{mk}_{ij\tau}$ (because the travel distance is smaller than $L^{mi}_{ij\tau}+L^{mk}_{ij\tau}$), while (51d) indicates that it leaves zone k at time $\tau^{mk}_{ij\tau}+1$ (because the travel distance is larger than $L^{mi}_{ij\tau}+L^{mk}_{ij\tau}$). Similarly, (51e) and (51f) ensure that a total of $g^m_{ij\tau}$ vehicles will arrive at their destinations in zone j within the interval $[\tau^{mj}_{ij\tau}, \tau^{mj}_{ij\tau}+1)$. Note that in a discrete-time model, this time interval suffices to inform the status transition of vehicles, i.e., from being en-route to being idle. Vehicles traveling along $p^m_{ij\tau}$ will become idle in zone j at time step $\tau^{mj}_{ij\tau}+1$. To capture this, we define another indicator as follows

$$\psi^{mt}_{ij\tau} = \begin{cases} 1 & \text{if } t = \tau^{mj}_{ij\tau}+1, \\ 0 & \text{otherwise.} \end{cases} \tag{52}$$

This allows us to characterize the number of vehicles completing trips and becoming idle in each zone at each time step. Correspondingly, the update equations of idle vehicles (24) and (25) can be rewritten as:

$$N^{I,t+1}_{i,A} = N^{I,t}_{i,A} - \sum_{j=1}^{N}\left(\frac{N^{I,t}_{i,A}}{N^{I,t}_{i,H}+N^{I,t}_{i,A}}\cdot\lambda^t_{ij} + f^t_{ij,A}\right) + \sum_{j=1}^{N}\sum_{\tau=1}^{T}\sum_{m}\psi^{mt}_{ji\tau}\phi^m_{ji\tau}g^A_{ji\tau}, \tag{53a}$$

$$N^{I,t+1}_{i,H} = N^{I,t}_{i,H} - \sum_{j=1}^{N}\left(\frac{N^{I,t}_{i,H}}{N^{I,t}_{i,H}+N^{I,t}_{i,A}}\cdot\lambda^t_{ij} + f^t_{ij,H}\right) - \hat{N}^t_{i,H} + \tilde{N}^t_{i,H} + \sum_{j=1}^{N}\sum_{\tau=1}^{T}\sum_{m}\psi^{mt}_{ji\tau}\phi^m_{ji\tau}g^H_{ji\tau}, \tag{53b}$$

where

$$g^A_{ji\tau} = \frac{N^{I,\tau}_{j,A}}{N^{I,\tau}_{j,A}+N^{I,\tau}_{j,H}}\cdot\lambda^\tau_{ji} + f^\tau_{ji,A}, \tag{54a}$$

$$g^H_{ji\tau} = \frac{N^{I,\tau}_{j,H}}{N^{I,\tau}_{j,A}+N^{I,\tau}_{j,H}}\cdot\lambda^\tau_{ji} + f^\tau_{ji,H}. \tag{54b}$$

Additionally, in the rest of the original model, e.g., (1)(7)(17)(19), the static travel time t_{ij} should be accordingly replaced by τ_{ijt}, which is dependent on the trip departure time. As such, we can extend the proposed mathematical model to characterize the

Table A.2
Trip time t_{ij}^{A} for AV on every OD pair (min).

O-D	Zone 1	Zone 2	Zone 3	Zone 4	Zone 5	Zone 6
Zone 1	8.23	12.17	14.79	23.93	25.19	29.29
Zone 2	11.85	8.57	13.21	16.09	21.12	27.46
Zone 3	14.40	12.63	7.92	21.48	14.48	21.84
Zone 4	21.28	14.96	20.55	11.79	15.15	19.30
Zone 5	23.81	19.63	13.62	14.43	9.70	14.88
Zone 6	27.60	26.1	21.39	18.20	14.51	10.74

time-varying travel time, taking into account the influence of ride-hailing vehicle movements on traffic congestion. It is important to note the extended model can be solved to obtain locally optimal solutions. However, the challenge of deriving globally optimal solutions for the extended model, or establishing theoretical guarantees regarding solution quality, is beyond the scope of this paper and remains a compelling avenue for future research.

7. Conclusions

This paper studies the optimal operational strategies of a platform that utilizes a combination of AVs and human drivers to provide mobility-on-demand services over a transportation network. We analyze a mixed fleet where AVs maintain a consistent fleet size under the central coordination of the TNC platform, while human drivers dynamically engage with the ride-hailing market, influenced by their anticipated earnings during different times. The complex strategic interactions among human drivers, passengers, and the TNC platform, which unfold across various spatial and temporal contexts, are encapsulated within a game-theoretic model. The two-player game devised is resolved through a best-response algorithm. To verify the global optimality of the derived solution in light of the platform's profit-maximization problem's non-convex nature, we introduce a novel convex relaxation approach, which provides an upper bound for the discrepancy between the candidate solution's objective value and the optimal global solution, establishing the best-response outcome as an ϵ-Nash equilibrium. Our developed model and solution methods are empirically validated using Manhattan's real-world data. Simulation results under different demand patterns (i.e., weekday and weekend) reveal that human drivers' spatial distribution closely mirrors real-time fluctuations in passenger demand, effectively catering to peak demand periods. In contrast, AV deployment aligns with the broader, predictable daily patterns of urban passenger flow across different city regions from day to night. We further compare the mixed fleet model with scenarios exclusively utilizing human drivers and AVs, respectively. The comparison shows that the mixed fleet model excels in balancing the supply and demand across different times and locations, thereby leading to shorter wait times and lower travel cost for ride-hailing passengers, as well as improved profit for TNC platform.

This work can be further extended from several direction. This include (a) developing a stochastic model considering the uncertainties of the future demand and supply in the ride-hailing market; (b) developing a hierarchical framework that jointly determines the long-term strategic decisions of the platform, as well as the short-term operational decisions of the platform; and (c) further enhancing our analysis by incorporating traffic congestion and/or investigating the impact of regulatory policies on the ride-hailing system with mixed fleet.

CRediT authorship contribution statement

Di Ao: Writing – original draft, Methodology, Investigation, Formal analysis. **Zhijie Lai:** Methodology, Investigation, Formal analysis. **Sen Li:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Trip time

See Tables A.2 and A.3.

Table A.3
Trip time t_{ij}^{H} for human driver on every OD pair (min).

O-D	Zone 1	Zone 2	Zone 3	Zone 4	Zone 5	Zone 6
Zone 1	7.41	10.95	13.31	21.53	22.67	26.36
Zone 2	10.67	7.71	11.89	14.48	19.01	24.72
Zone 3	12.96	11.36	7.13	19.33	13.03	19.65
Zone 4	19.15	13.46	18.49	10.61	13.63	17.37
Zone 5	21.42	17.67	12.26	12.99	8.73	13.39
Zone 6	24.84	23.51	19.25	16.38	13.06	9.67

Table B.4
List of abbreviations.

Abbreviation	Definition
MoD	Mobility on demand
AMoD	Autonomous mobility on demand
AV	Autonomous vehicle
TNC	Transportation network company
MPC	Model predictive control
OEM	Original equipment manufacture
OD	Orientation-Destination
TLC	Taxi & Limousine Commission

Appendix B. Abbreviations

See Table B.4.

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